UNIVERSITY OF LEEDS

Transformers

Deep Learning - COMP5203M

2022-23

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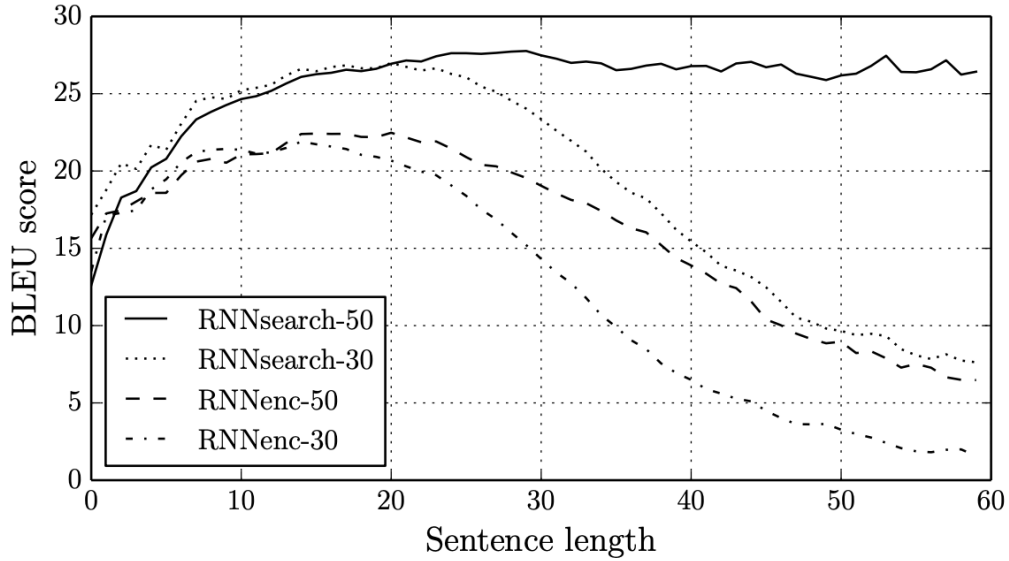
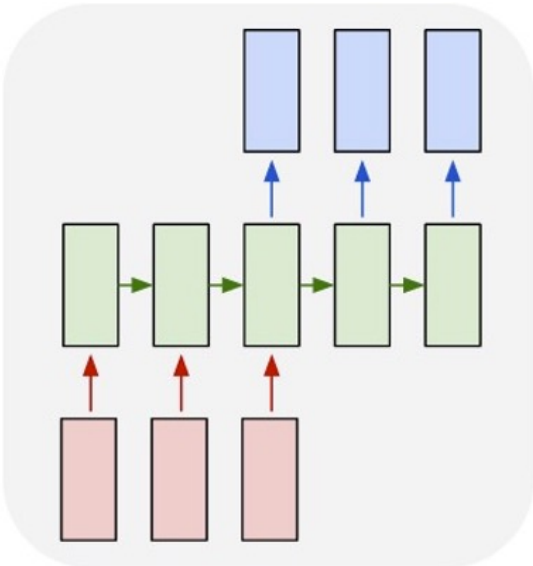
Limitations of RNN

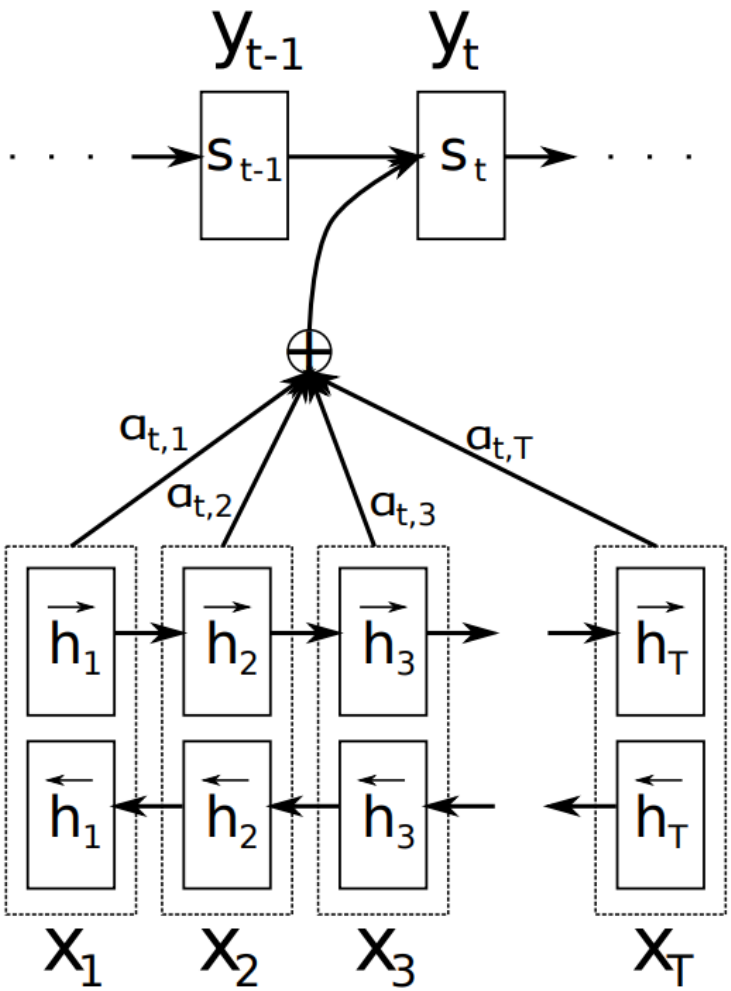
1. Dealing with long sentences or even paragraphs. Forgetting long-range dependencies.
2. Vanishing gradients and hard to train.
3. Sequential processing, therefore, hard to parallelise.
4. Having to process the entire input before generating output.





Bahdanau, Dzmitry, Kyunghyun Cho, and Yoshua Bengio. "Neural machine translation by jointly learning to align and translate." *arXiv preprint arXiv:1409.0473* (2014)

RNN with Attention



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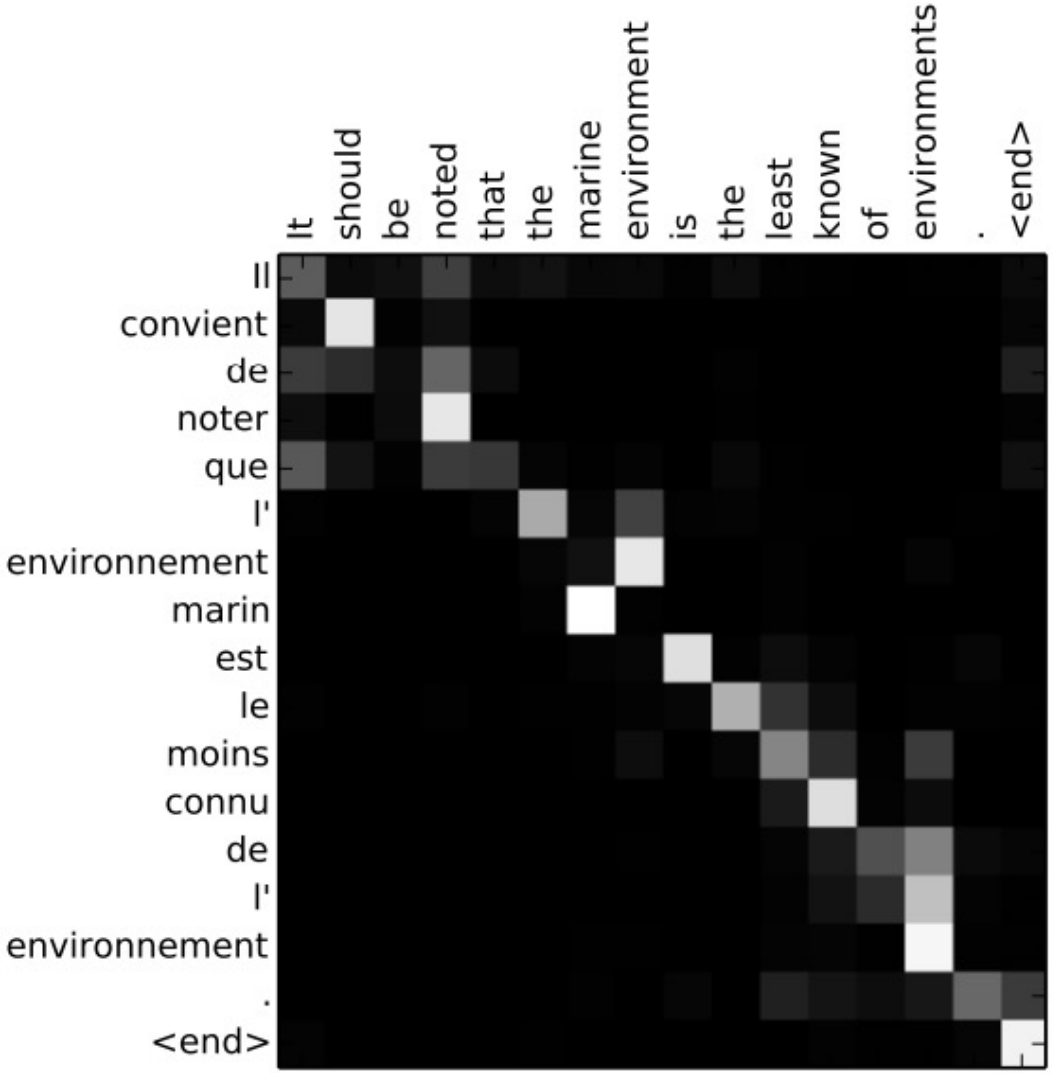
Attention example



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Transformers

- The attention mechanism is an important contribution to NLP.
  - But RNN architecture is a major bottleneck.
  - RNNs are difficult to parallelise.
- Solution? Getting rid of RNN altogether.
- Transformers core idea: Attention is all you need!

Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Transformers

Core ideas:

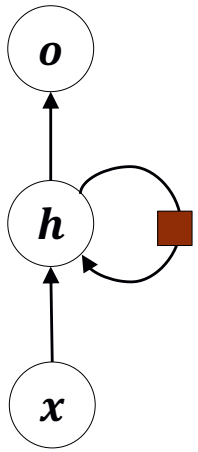
1. Positional encoding
2. Attention
3. Self-attention

Positional encoding (brief)

- In RNN positional information is encoded into the architecture by means of sequential processing.
- Transformers disentangle this by removing word order from the architecture and moves it to data.
- A simple example (*for illustration*):

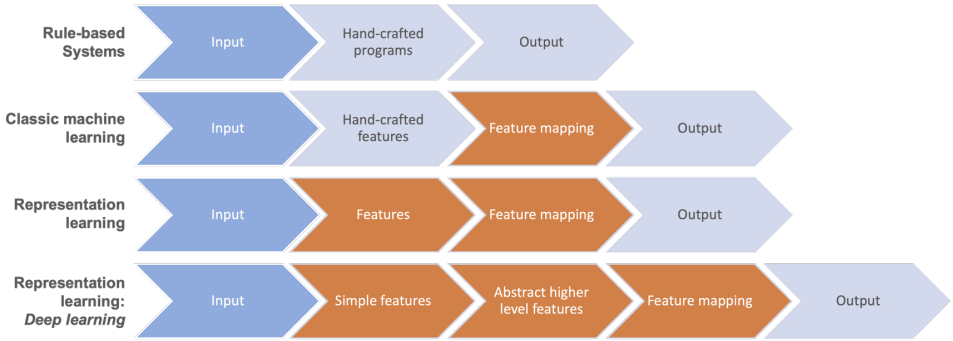
  (“Deep”, 1), (“learning”, 2), (“is”, 3), (“cool”, 4)

- The network learns to use these positional information.
- What’s the benefit? parallelism.



Self-attention (brief)

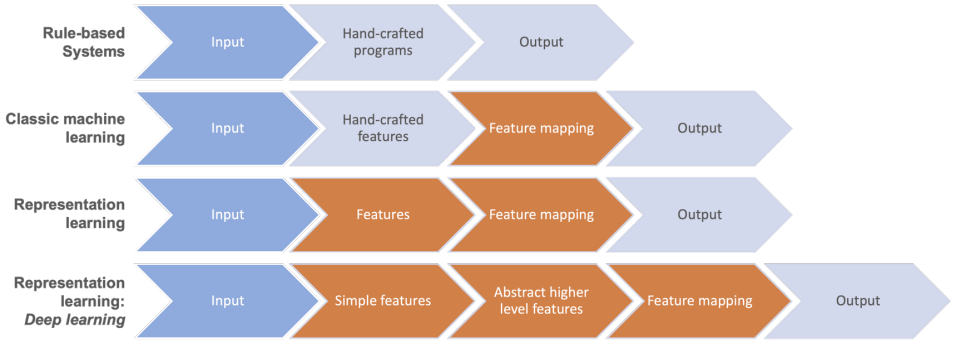
- The idea of deep learning is to be able to build powerful internal representations.

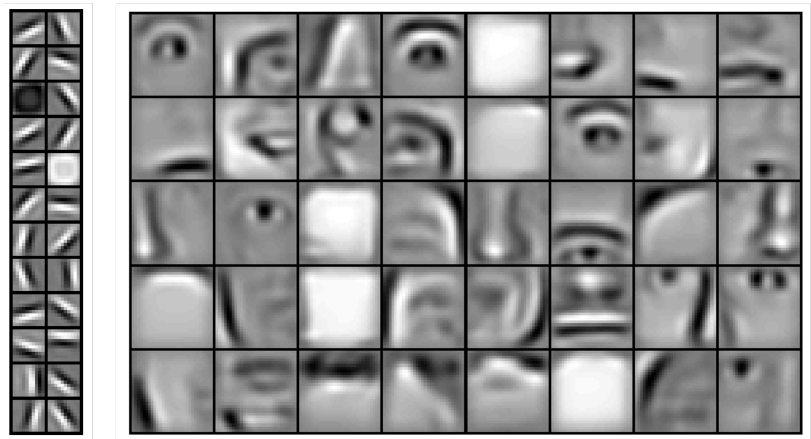


https://en.wikipedia.org/wiki/Grammar

Self-attention (brief)

- The idea of deep learning is to be able to build powerful internal representations.
- In CNN: convolution operator → edge detection → higher level features.





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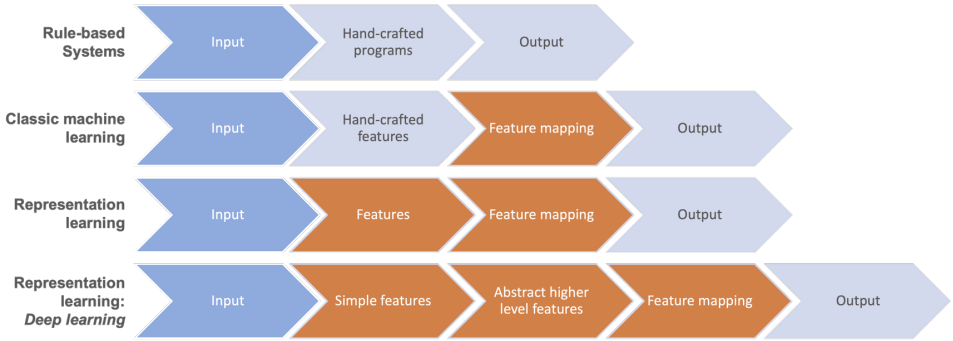
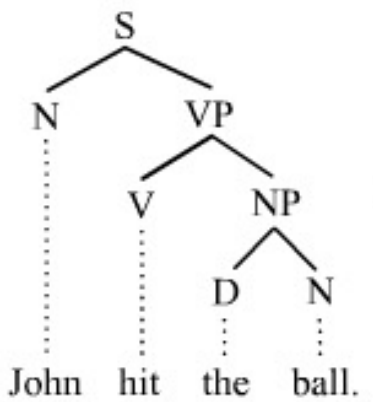
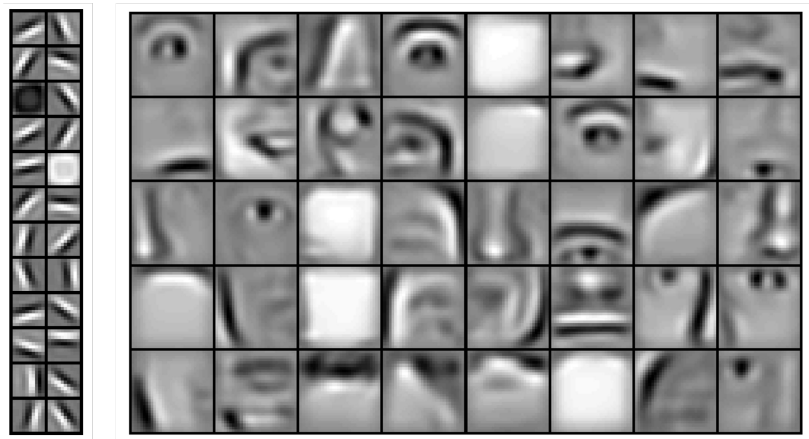
Self-attention (brief)

- The idea of deep learning is to be able to build powerful internal representations.
- In CNN: convolution operator → edge detection → higher level features.
- In language models: automatically learn parts of speech, rules of grammar, synonymous, etc.





Constituency-based parse tree



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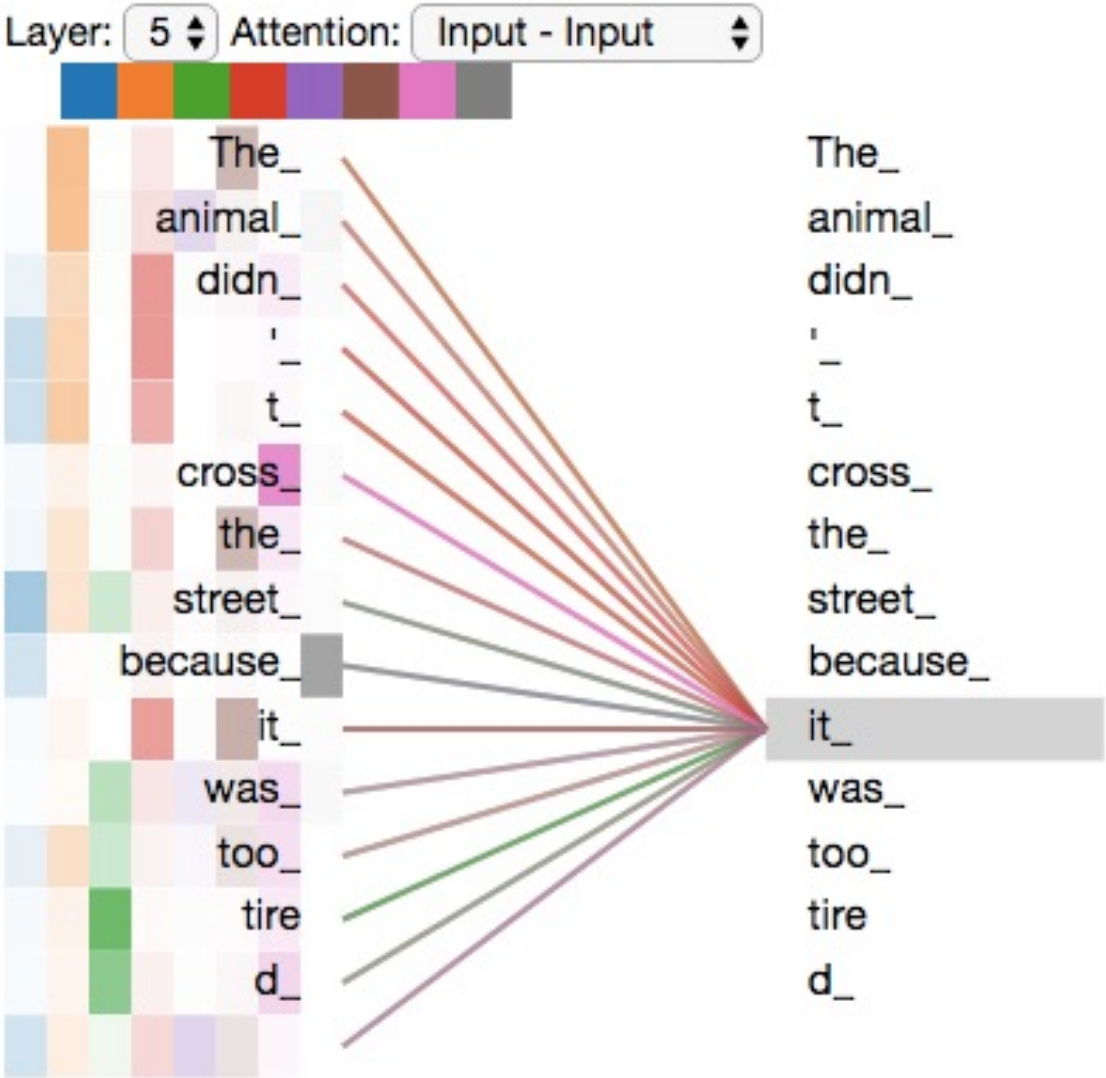
Self-attention (brief)

- Self-attention: applying attention mechanism to the input itself.

- Self-attention helps neural networks to build more powerful representations.



https://colab.research.google.com/github/tensorflow/tensor2tensor/blob/master/tensor2tensor/notebooks/hello_t2t.ipynb

Transformers

Core ideas:

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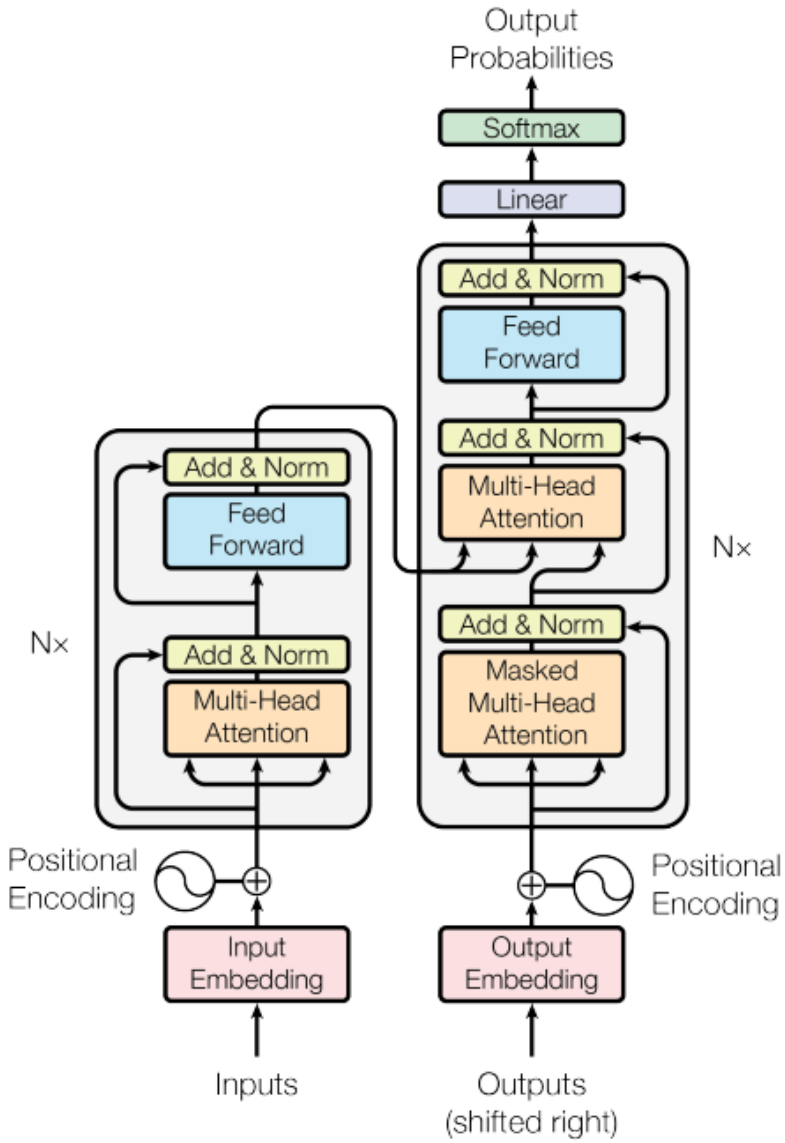
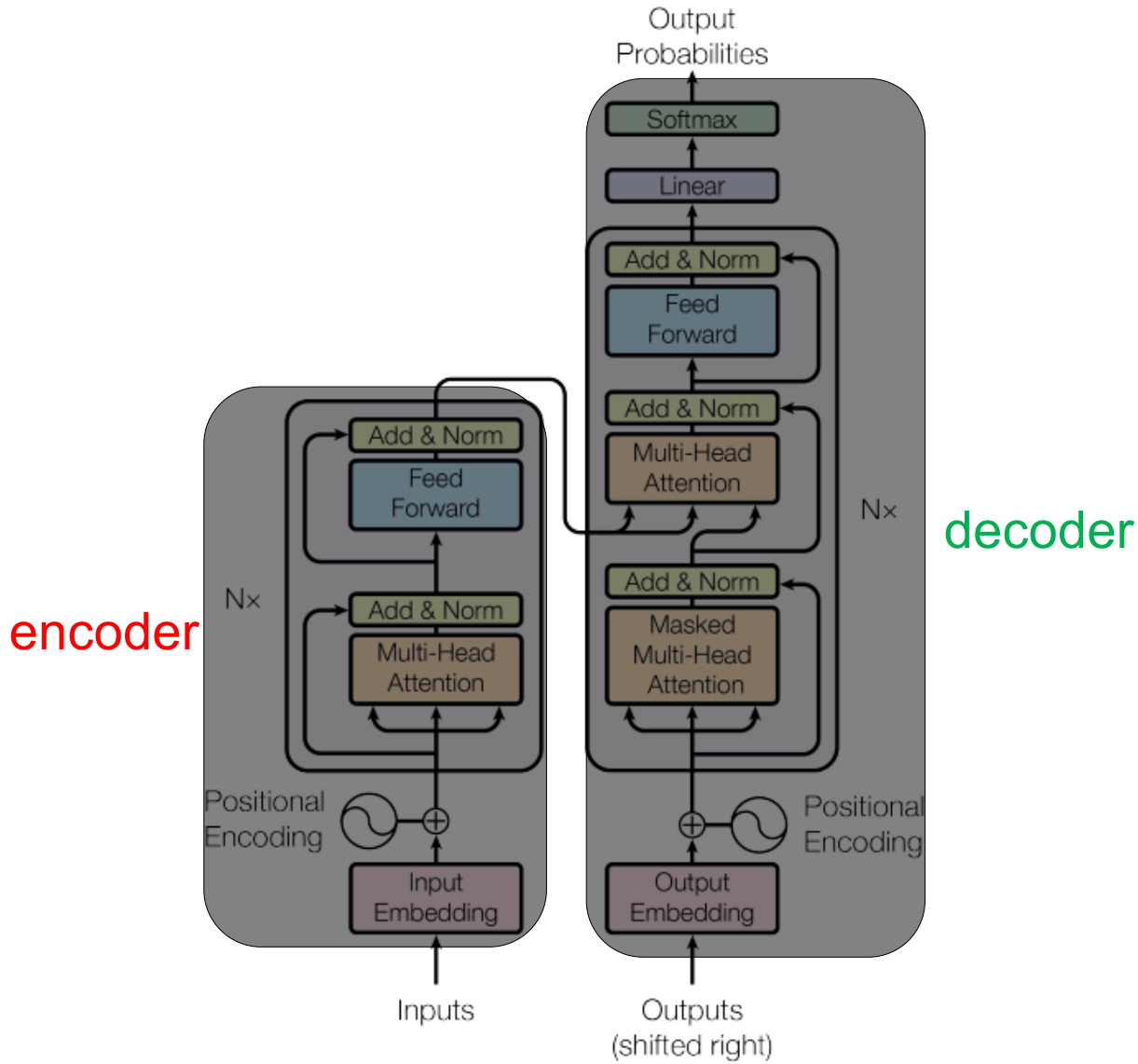


Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Transformer: bird-eye view



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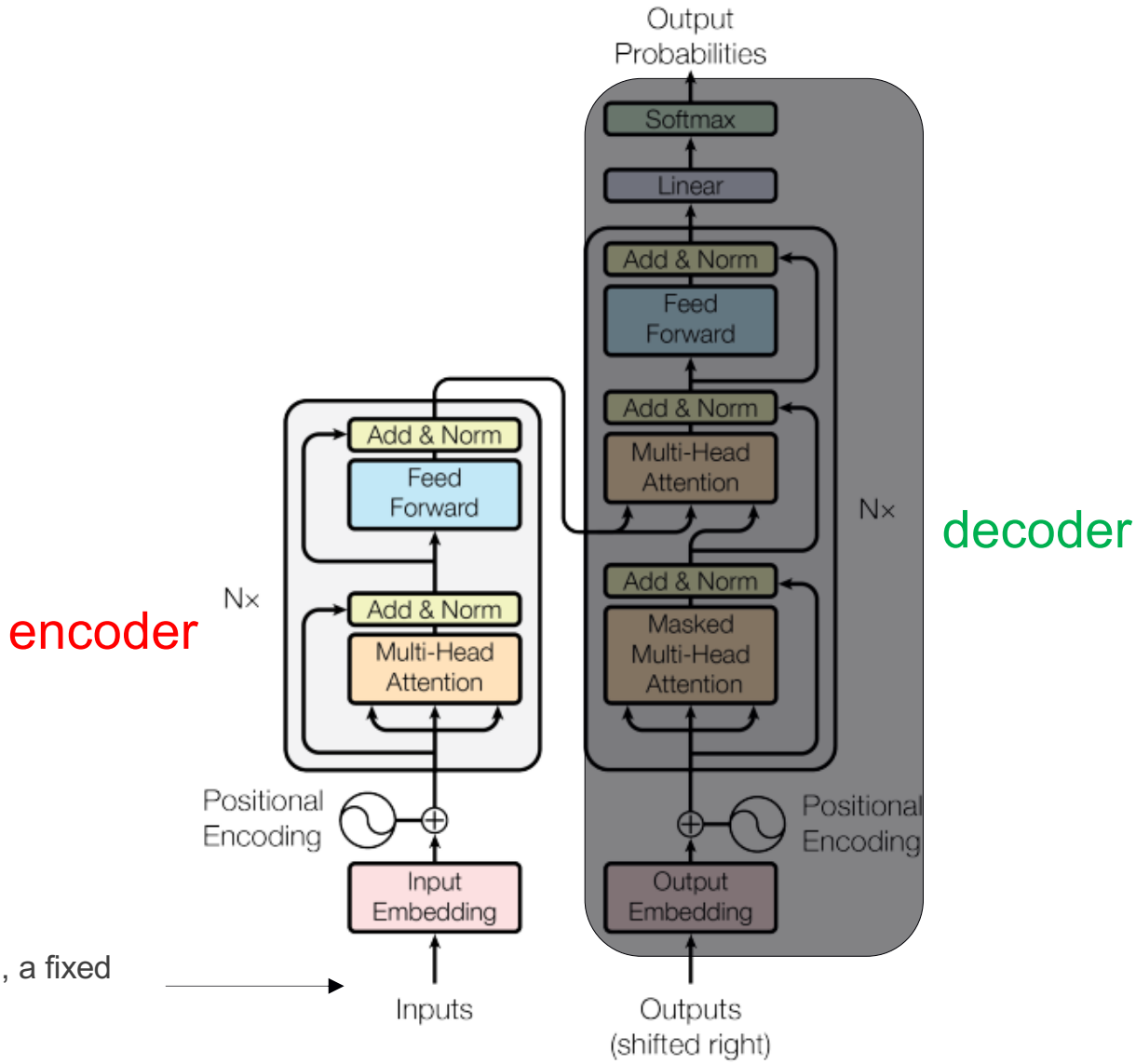
Transformer: Encoder



Input size is fixed, i.e., a fixed number of tokens.

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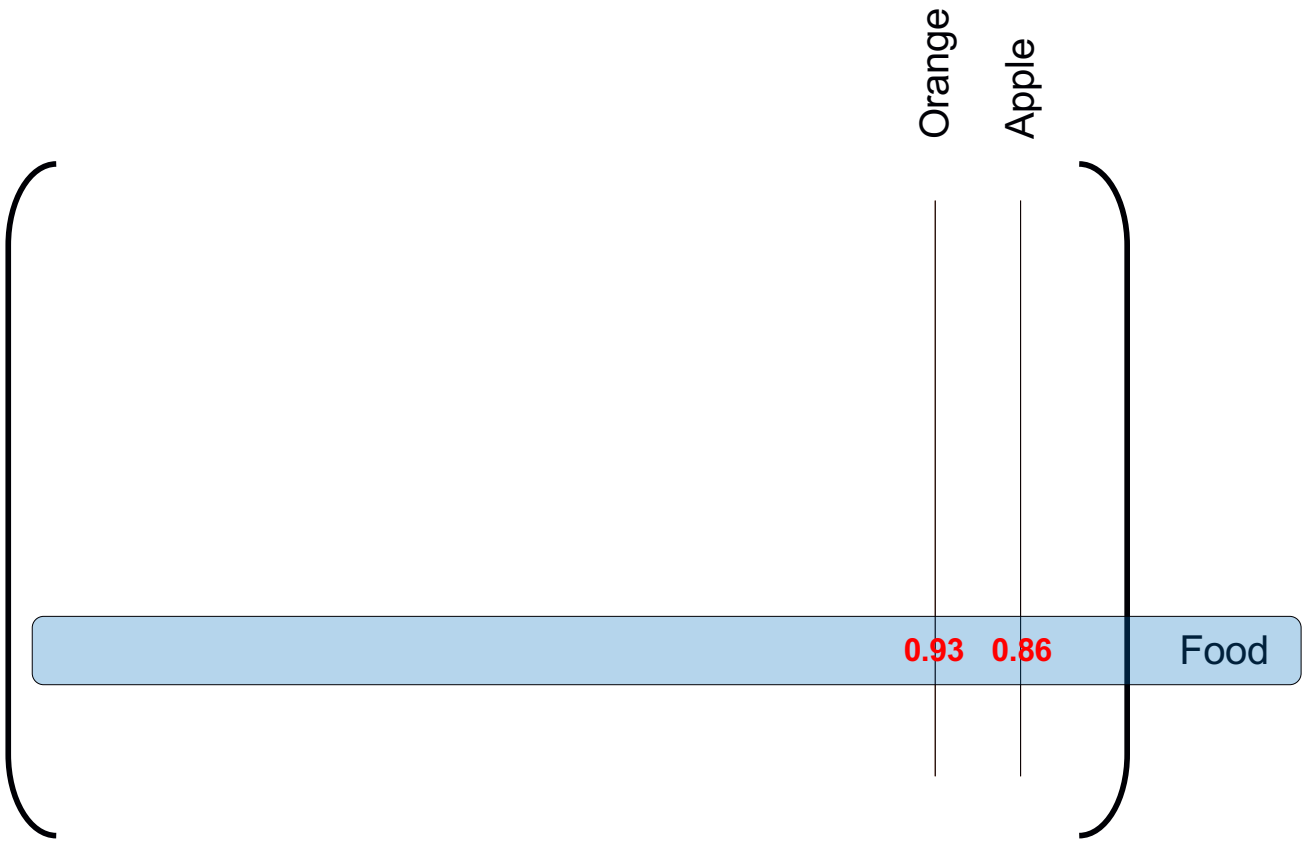
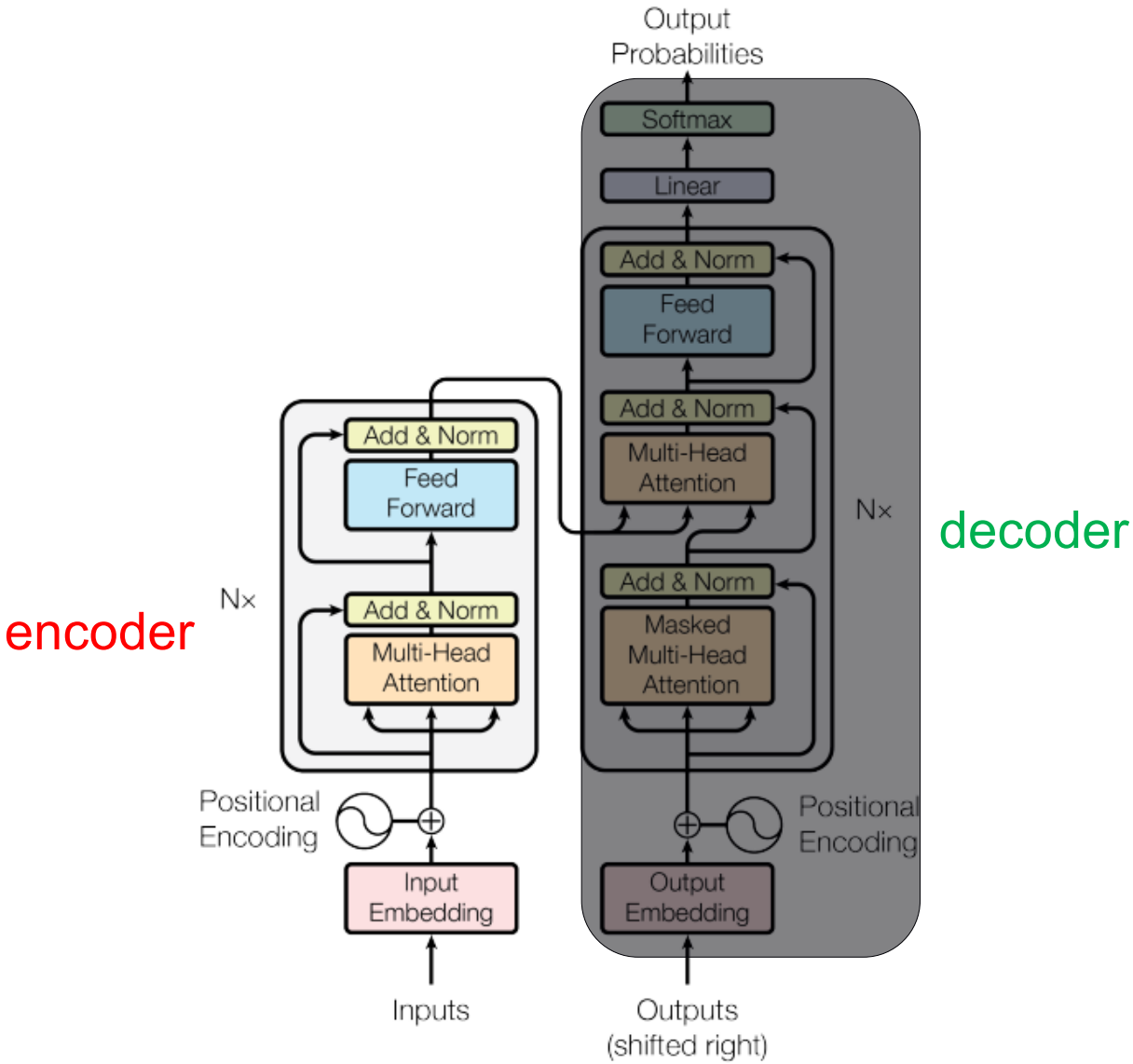
Word Embedding

$$\mathbf{E}_{d\times|V|} =$$



Encoder



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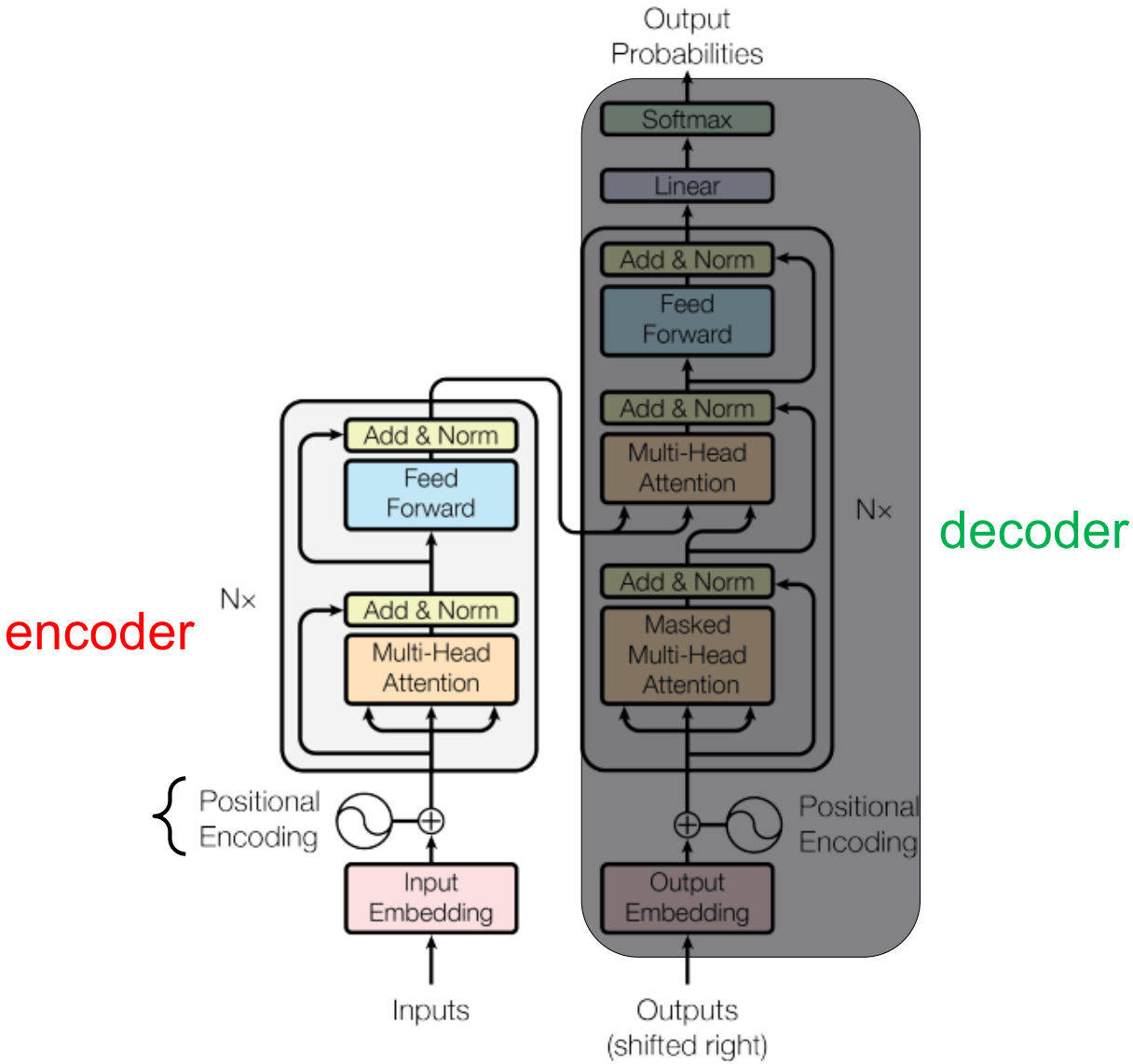

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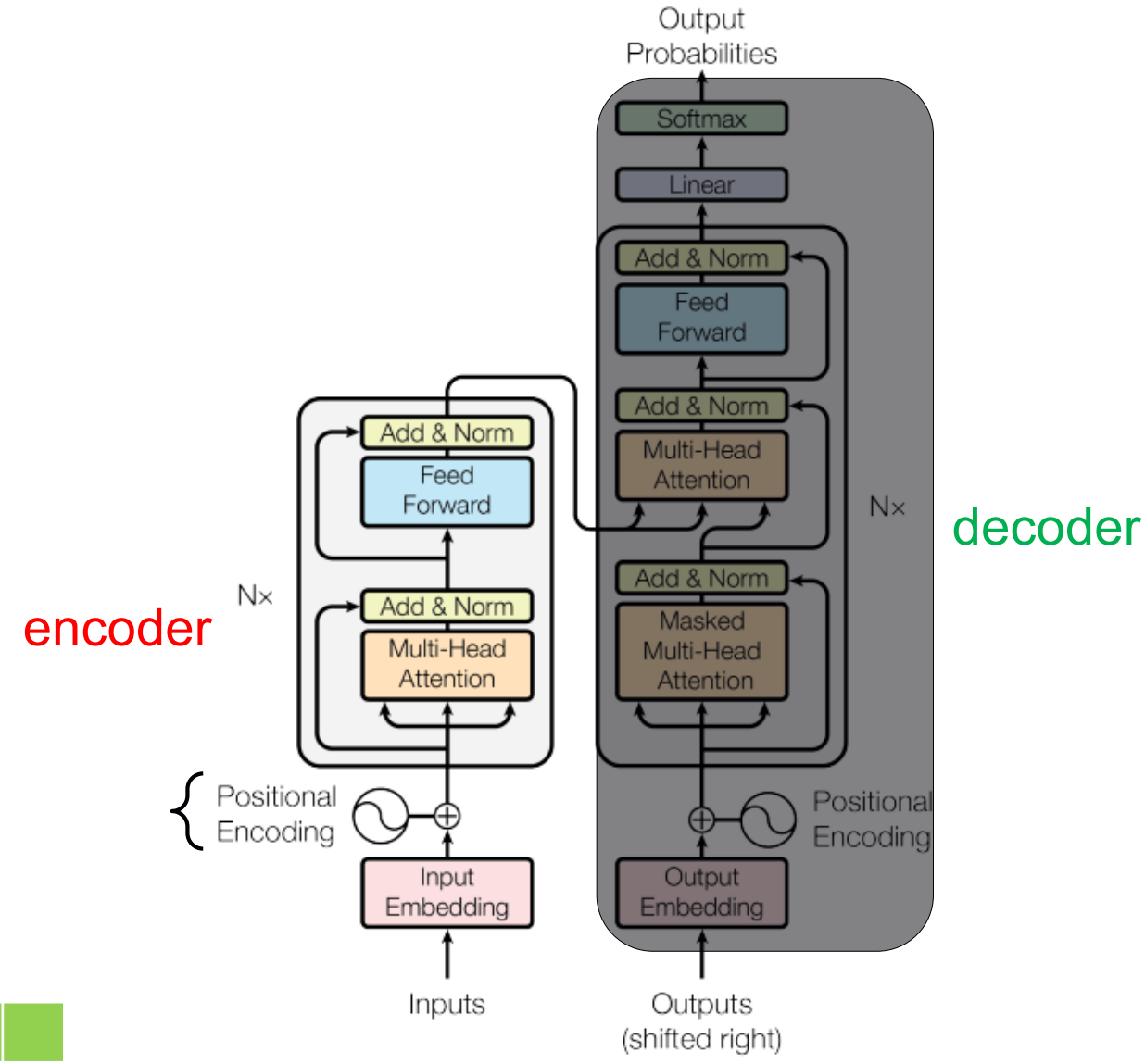

$x^{<i>}$

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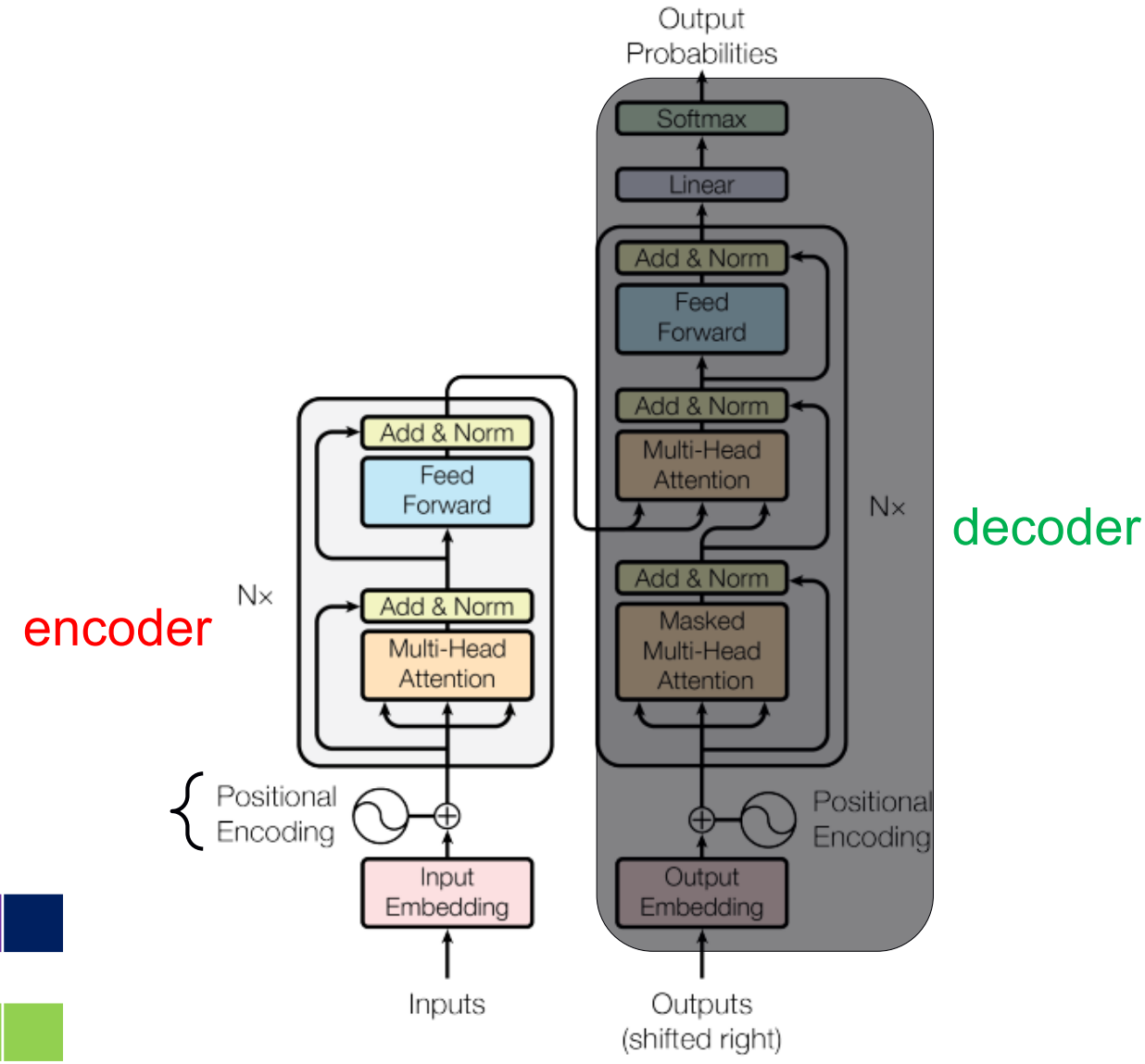

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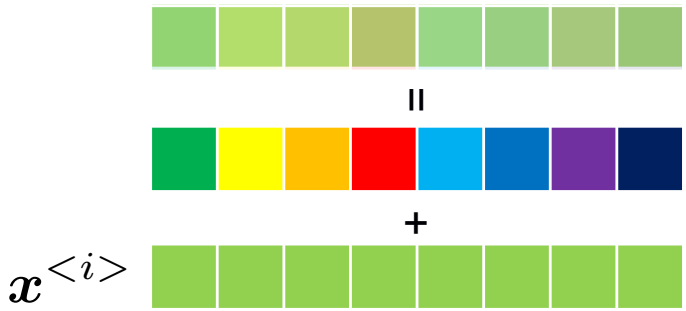

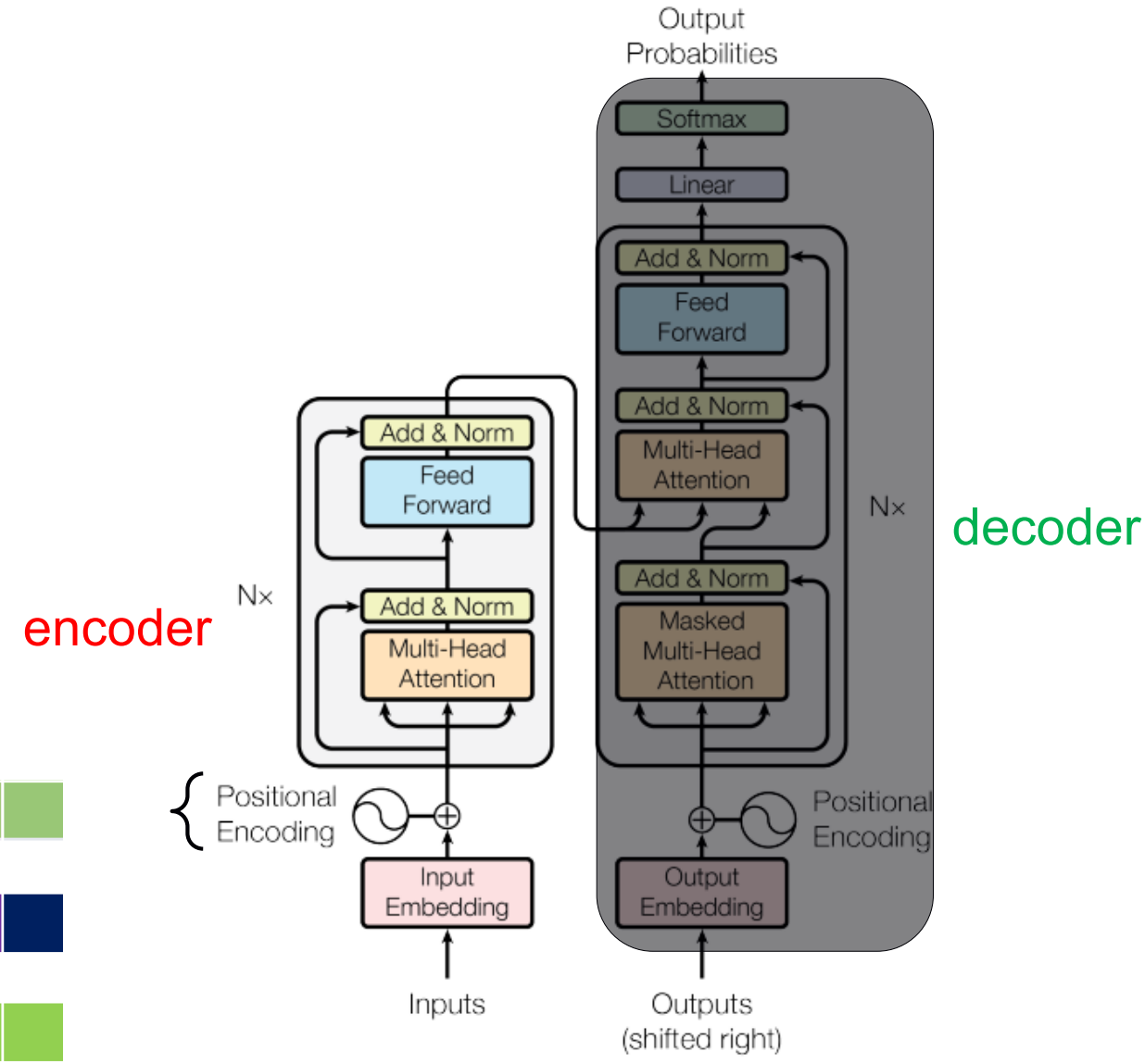

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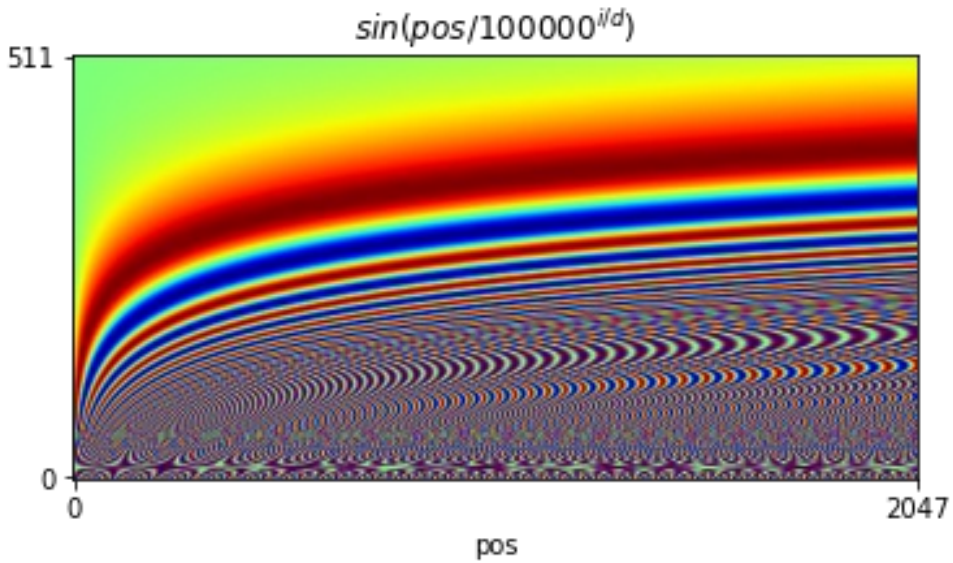
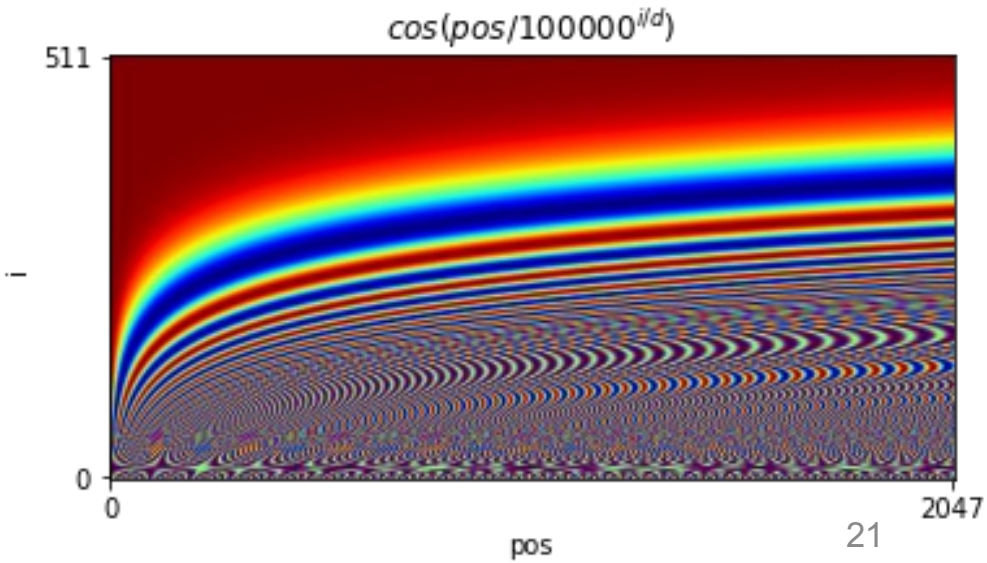
Position encoding

The <u>sequence</u> of tokens is unused by the attention and fully connected layers. The tokens are treated as a *bag* and not a sequence.

To introduce temporal encoding, add vectors encoding position to the embedded tokens.

$$pe(pos, i) = \begin{cases} sin\left(pos/10000^{2i/d}\right) & i \text{ even} \\ cos\left(pos/10000^{2i/d}\right) & i \text{ odd} \end{cases}$$





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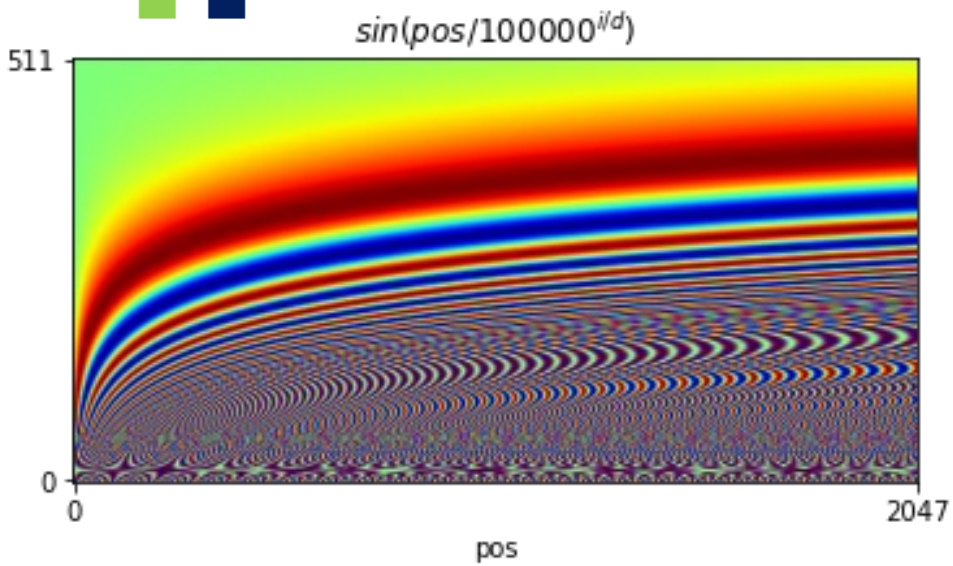

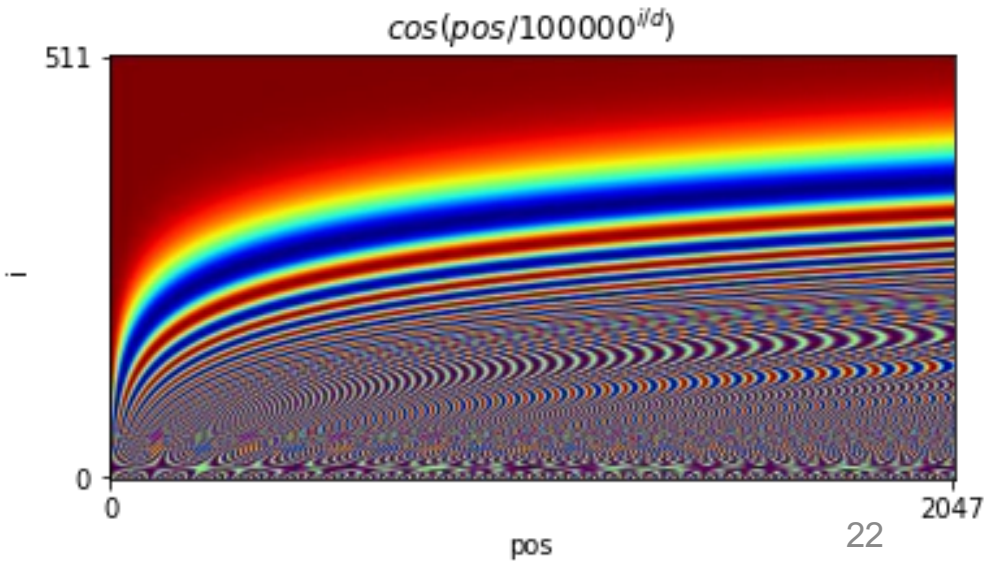

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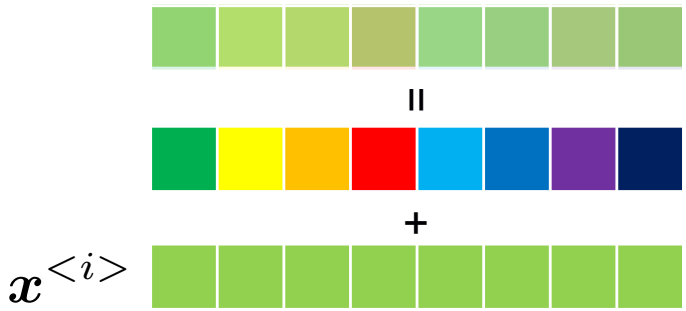

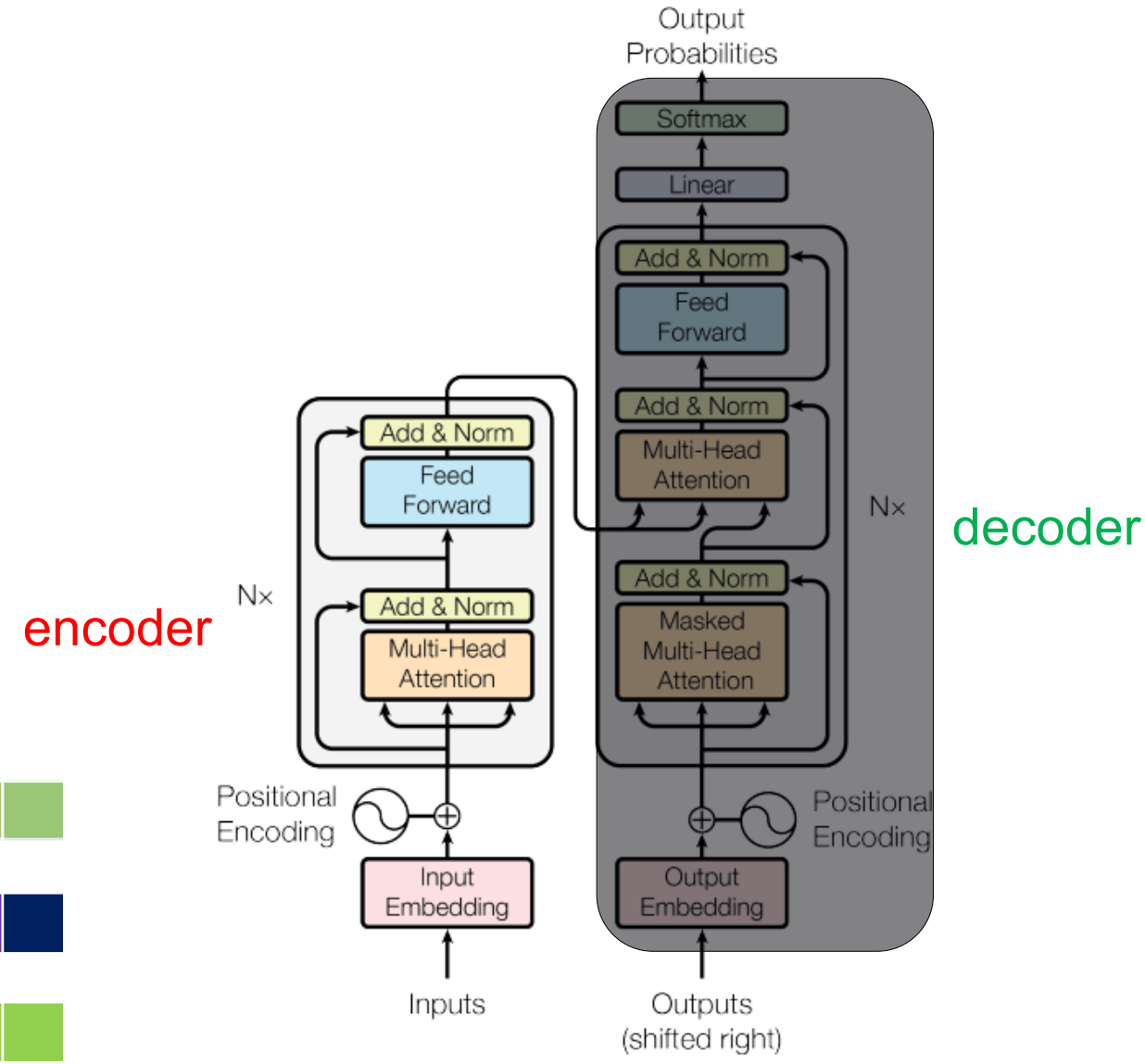

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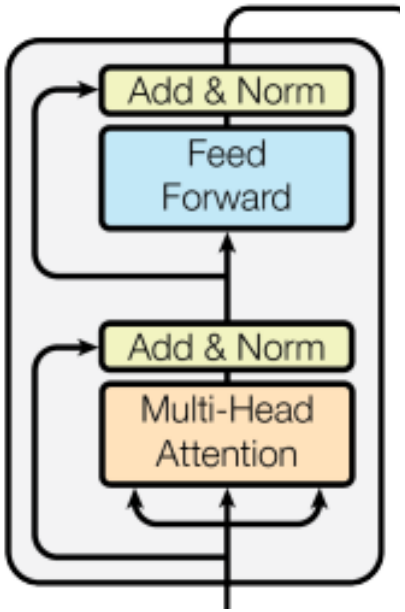

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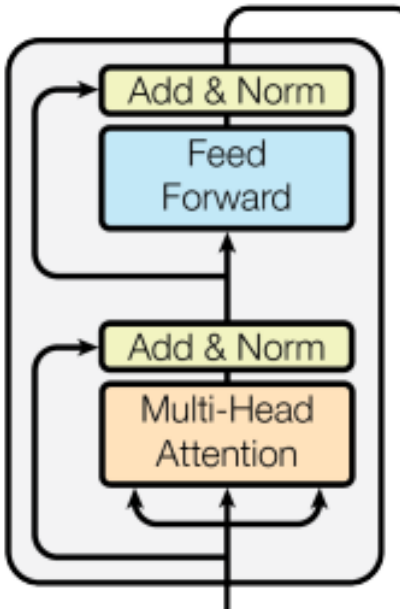

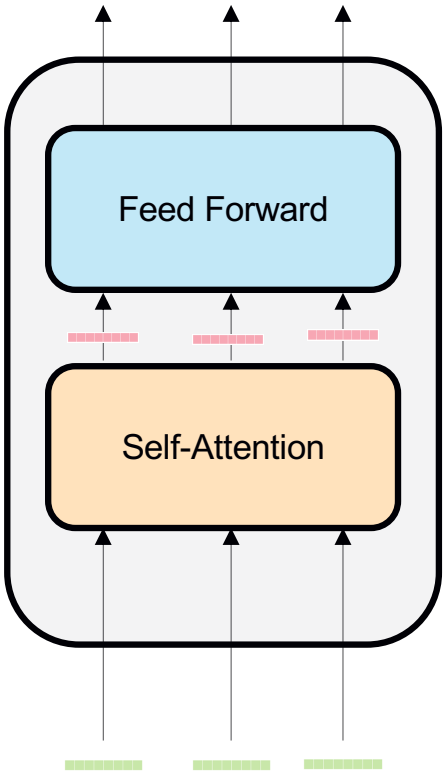

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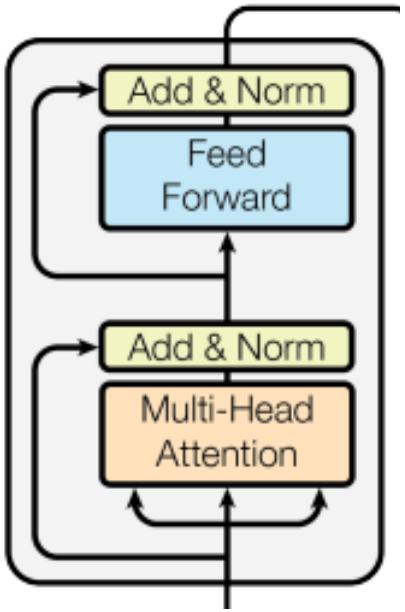

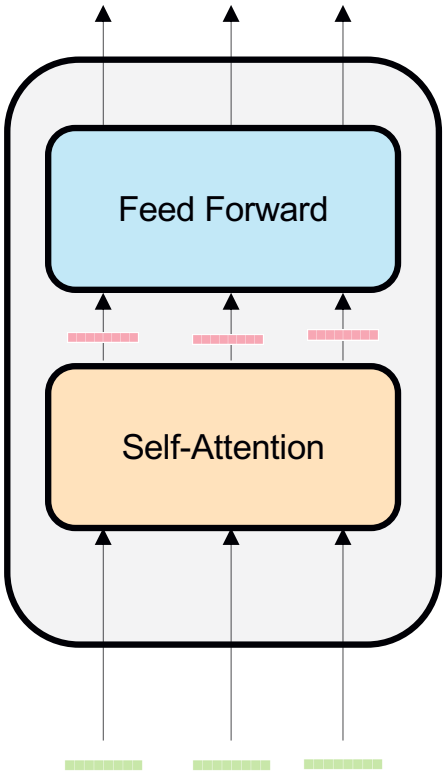

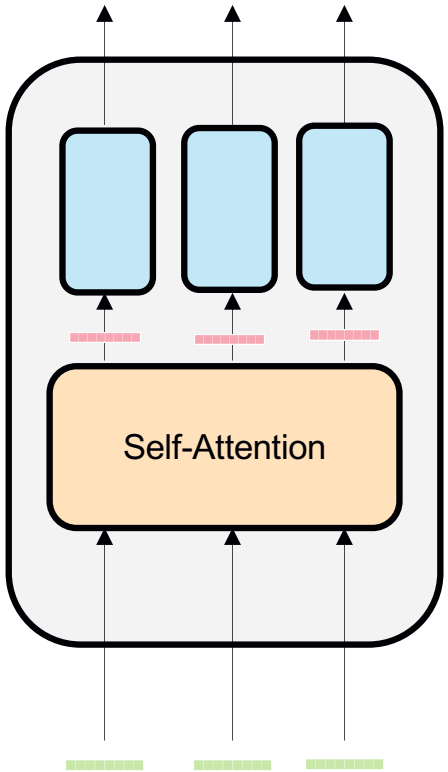

Parallelism

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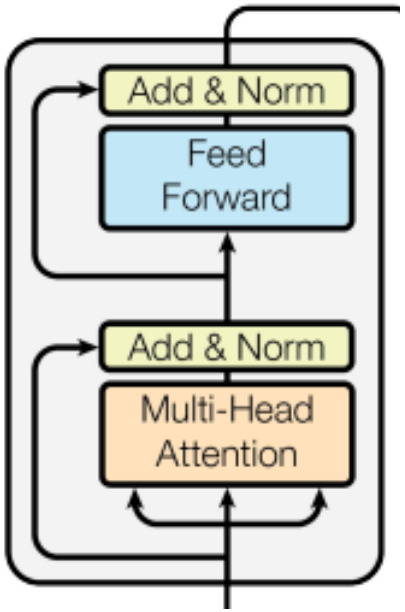
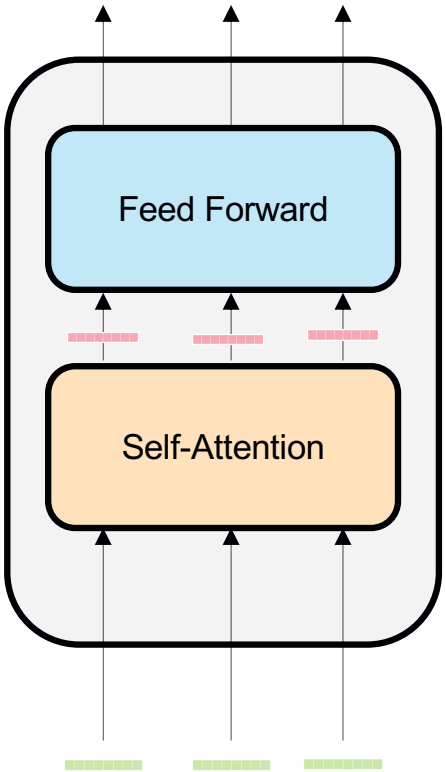
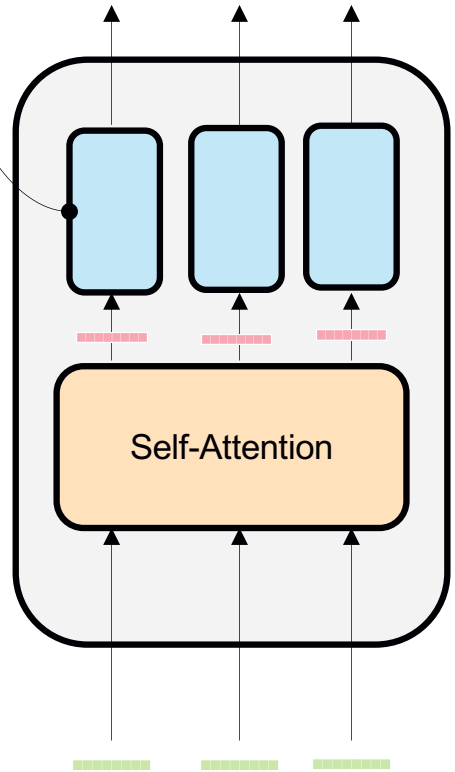
Encoder





Two fully connected layers with ReLU. Same parameters used for all positions (tokens).



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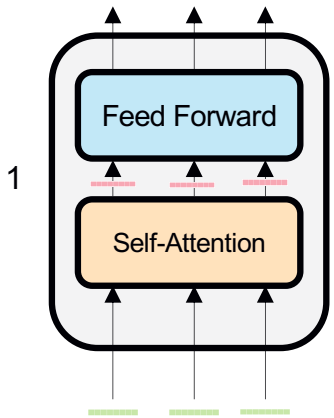

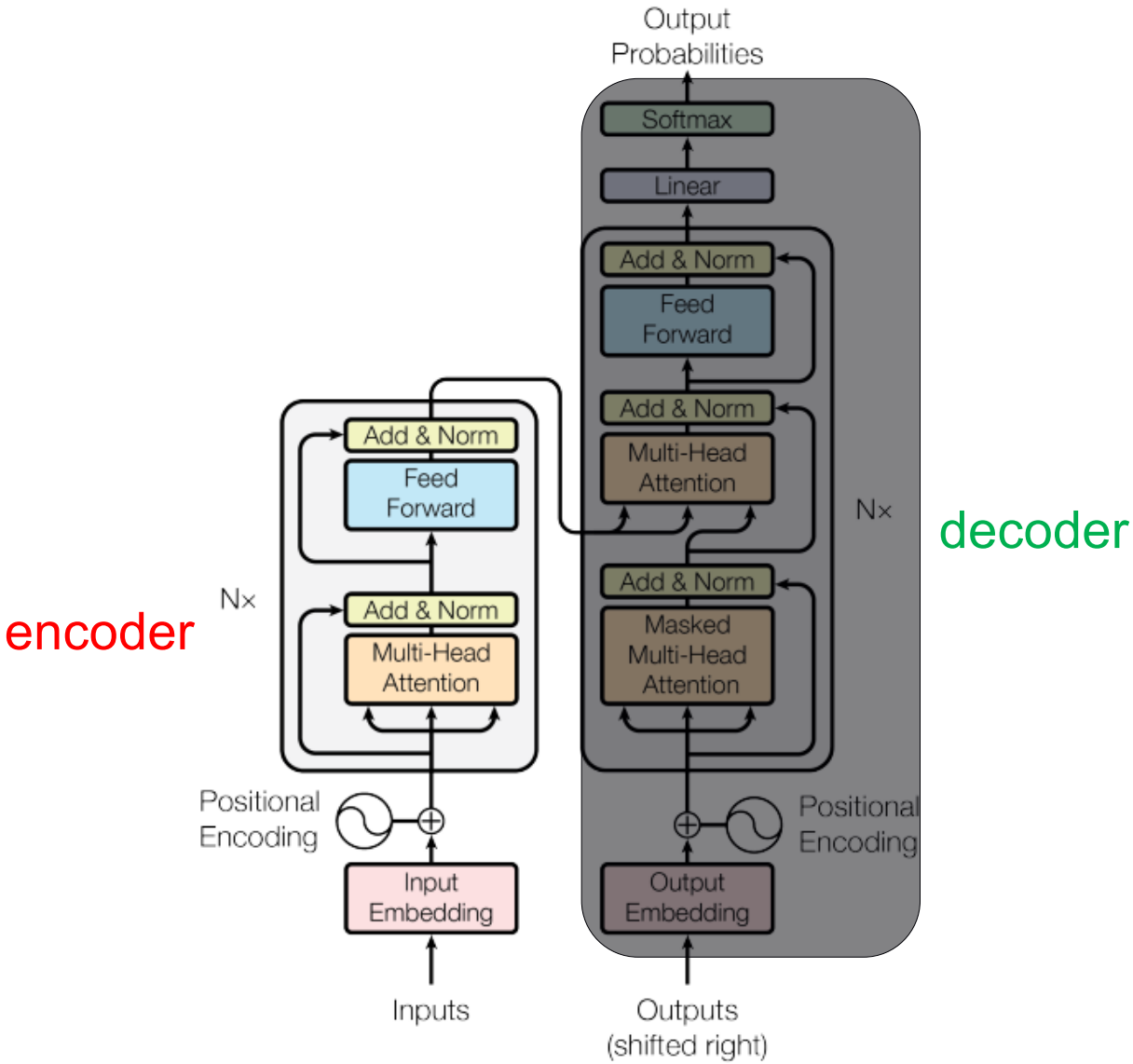

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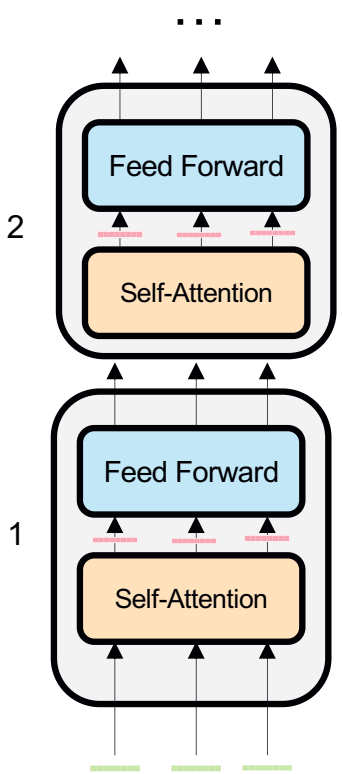

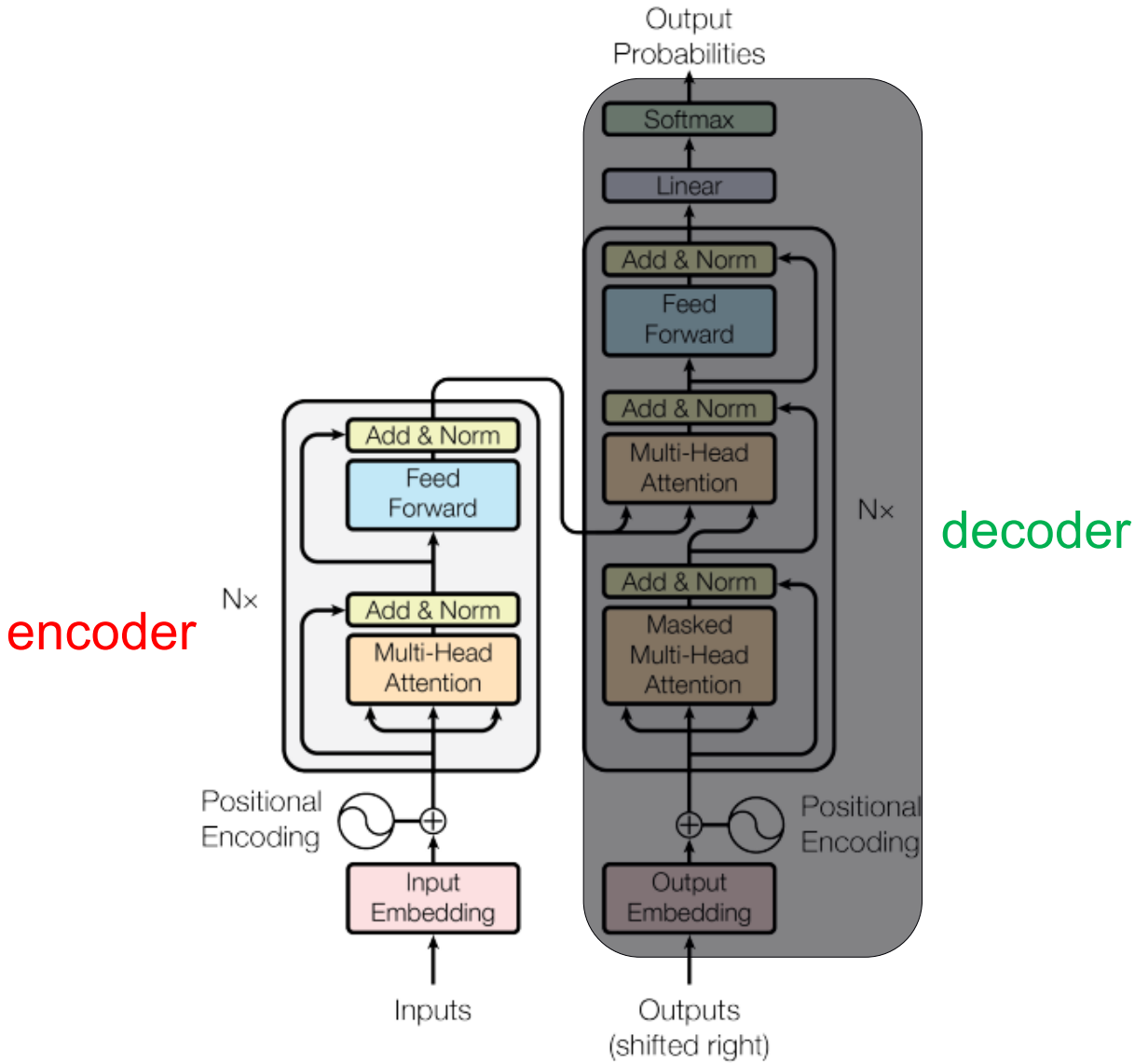

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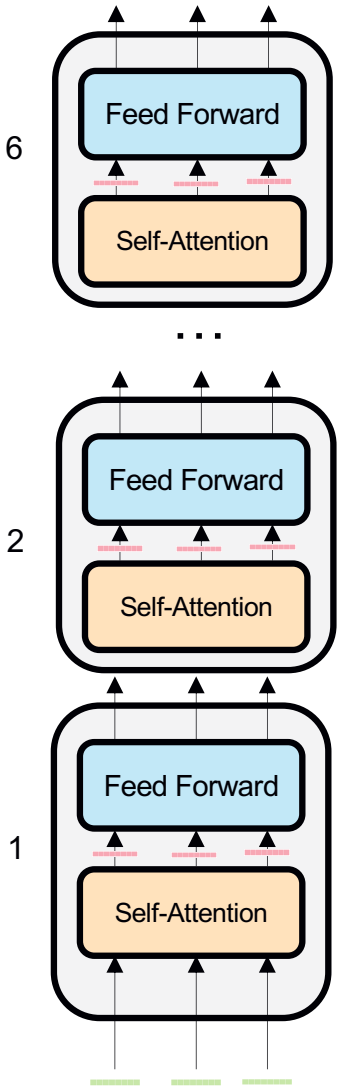
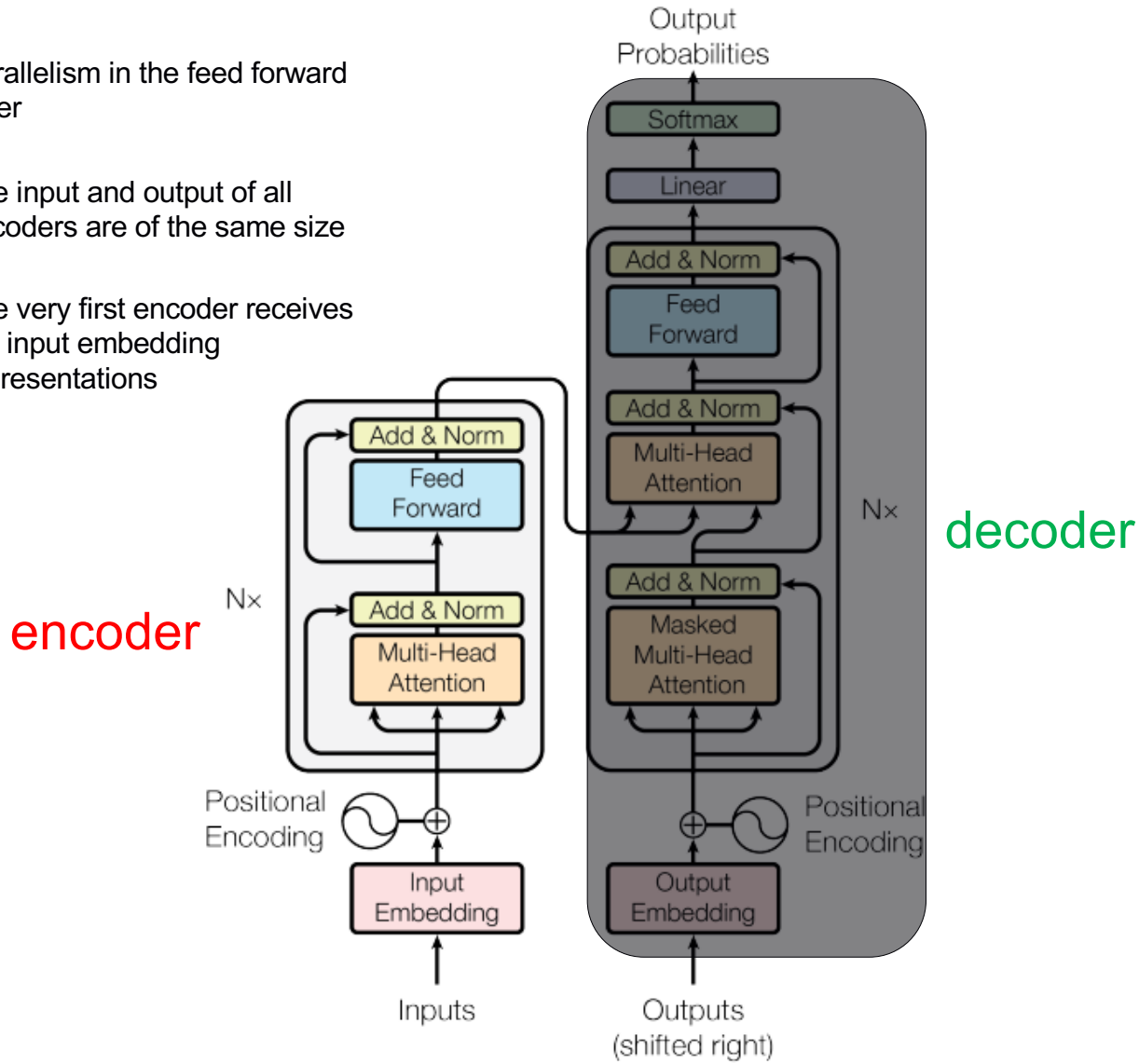


- Parallelism in the feed forward layer
- The input and output of all encoders are of the same size
- The very first encoder receives the input embedding representations



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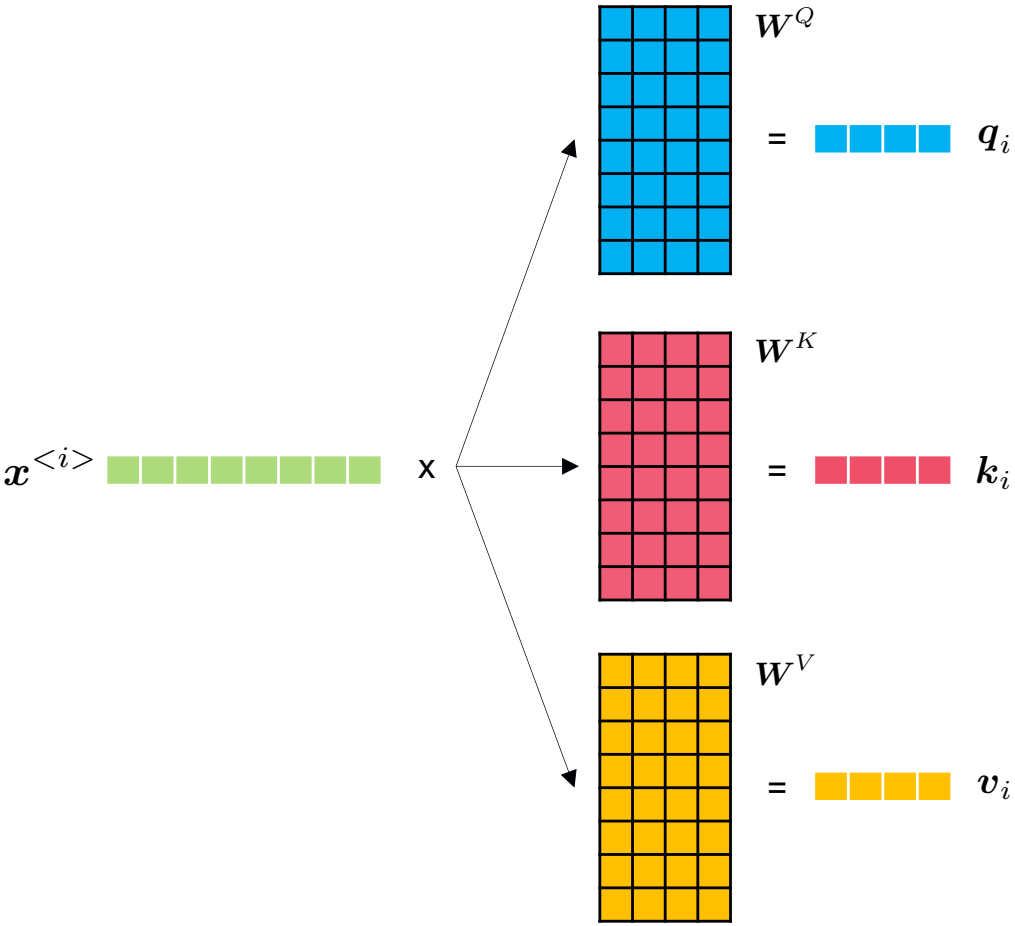
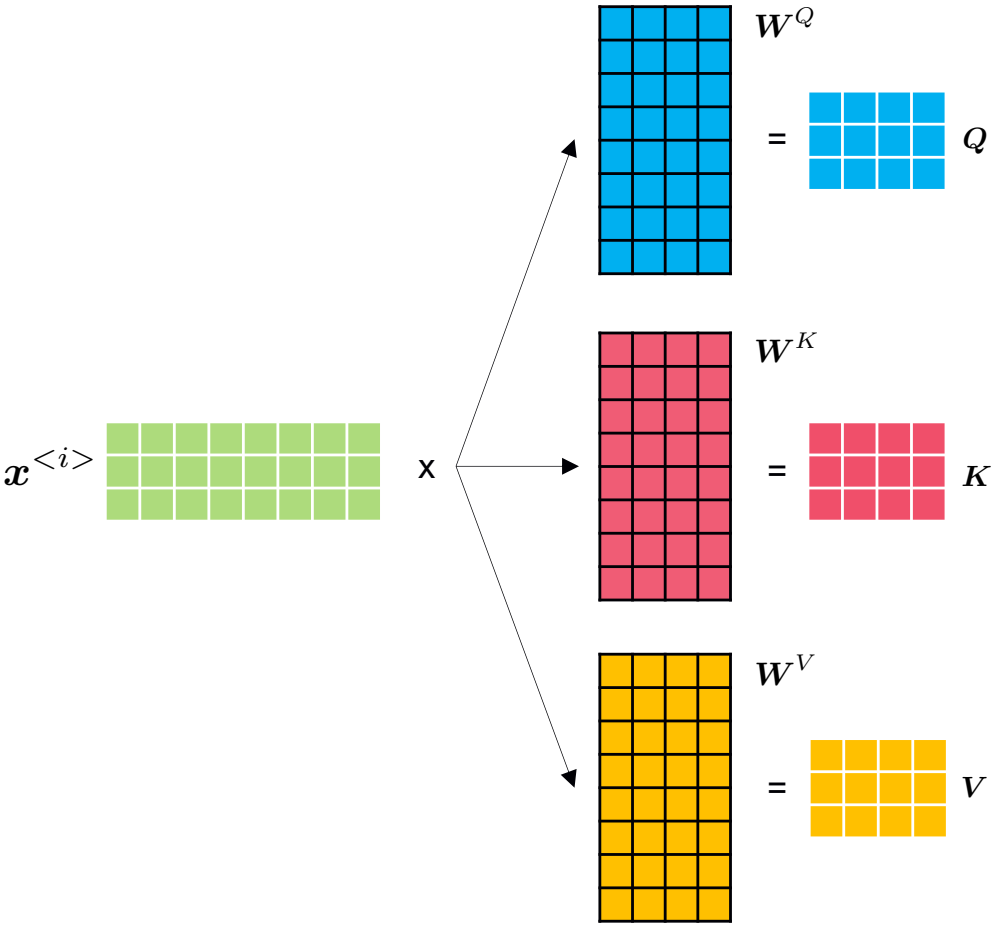
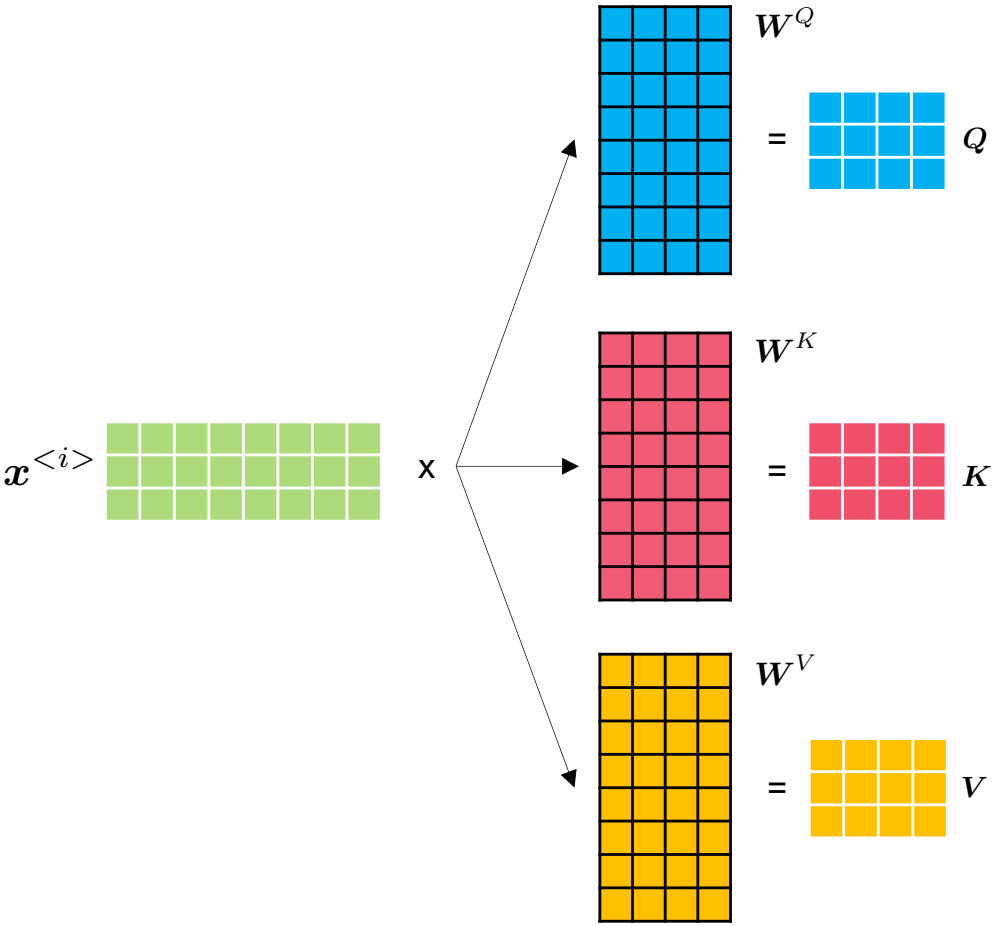
Self-attention: single token



Self-attention: matrix form



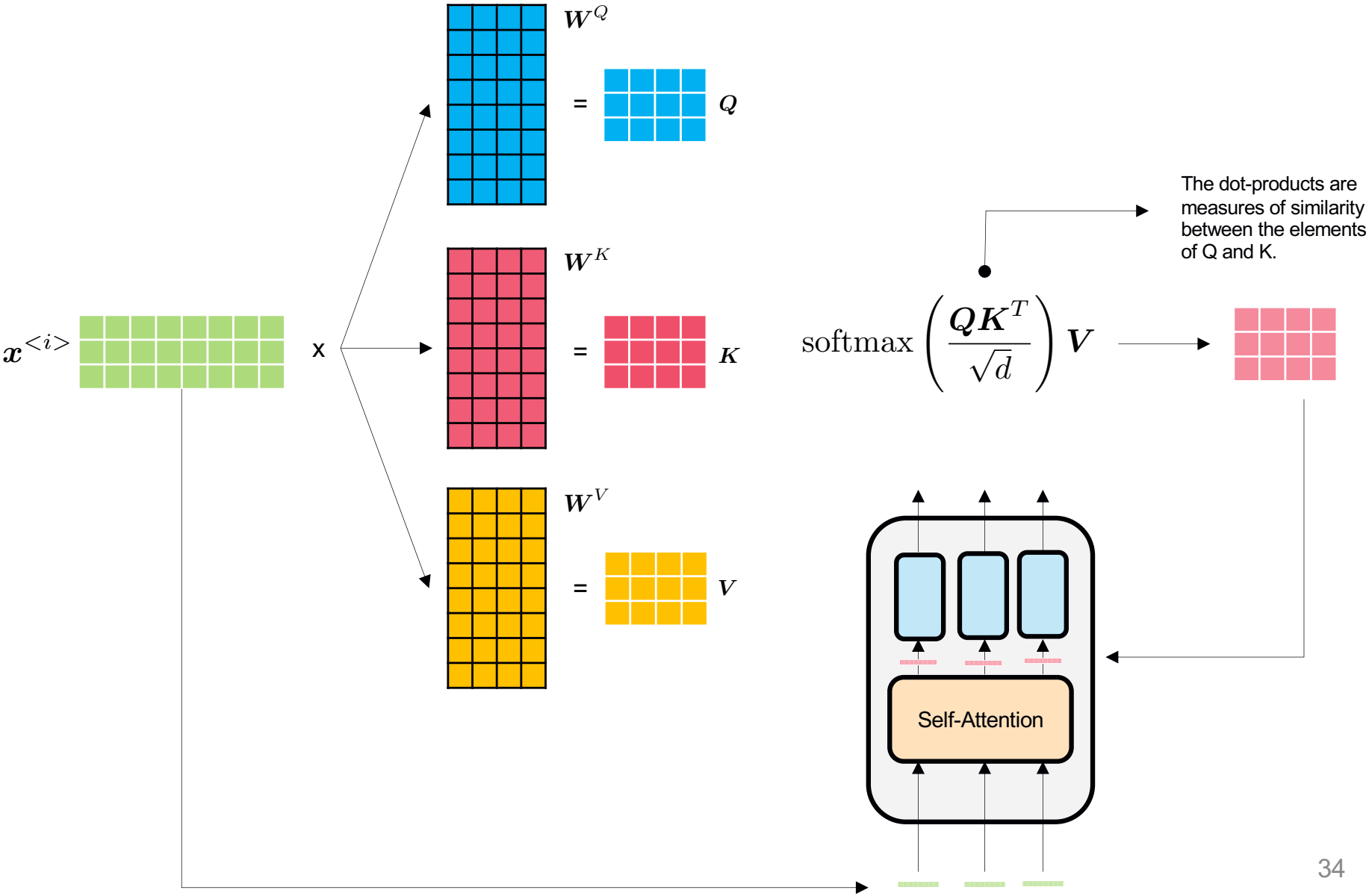
Self-attention: matrix form



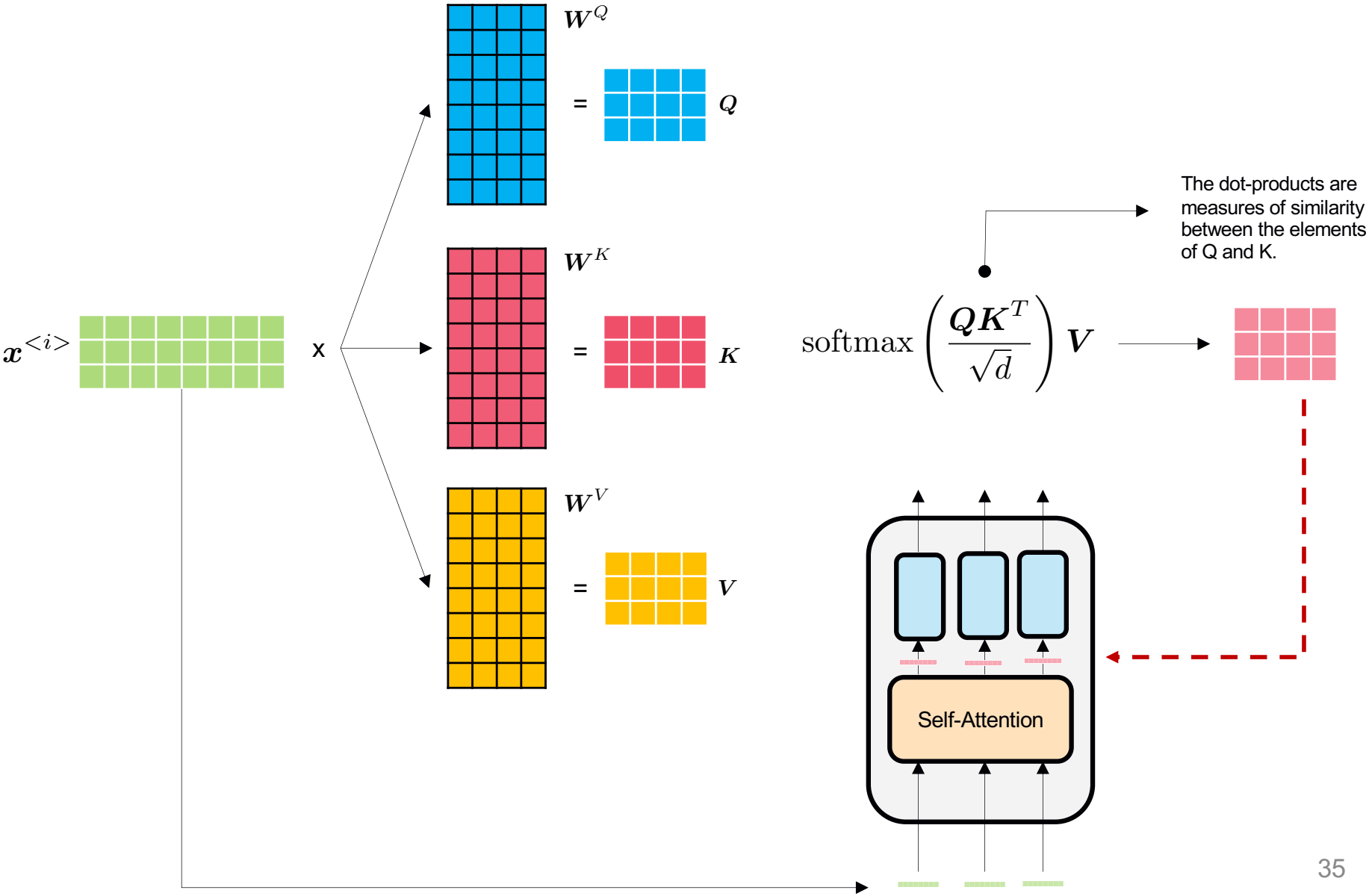
The dot-products are measures of similarity between the elements of Q and K.

$$\text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d}}\right)\boldsymbol{V}$$

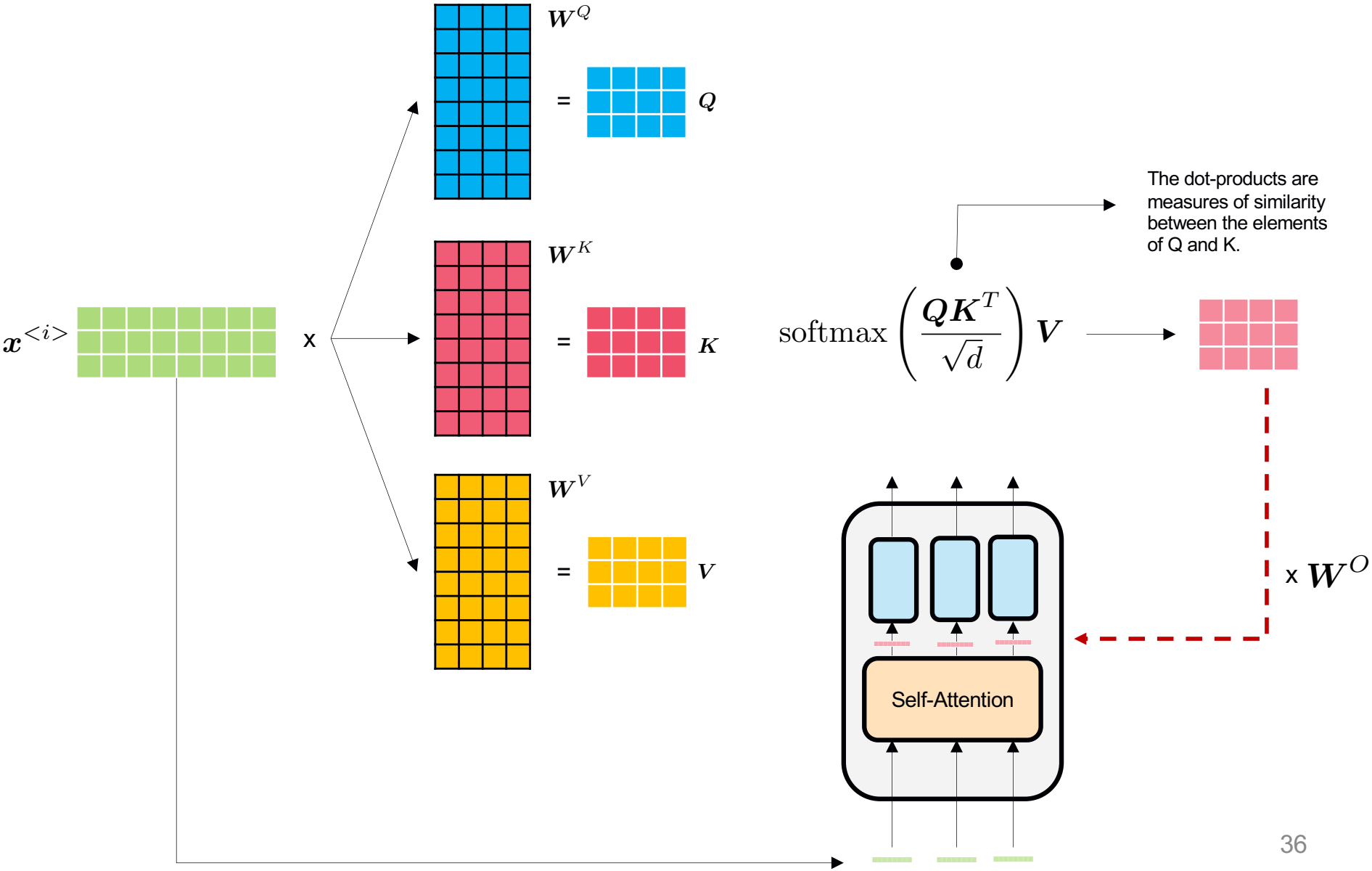
Self-attention: matrix form



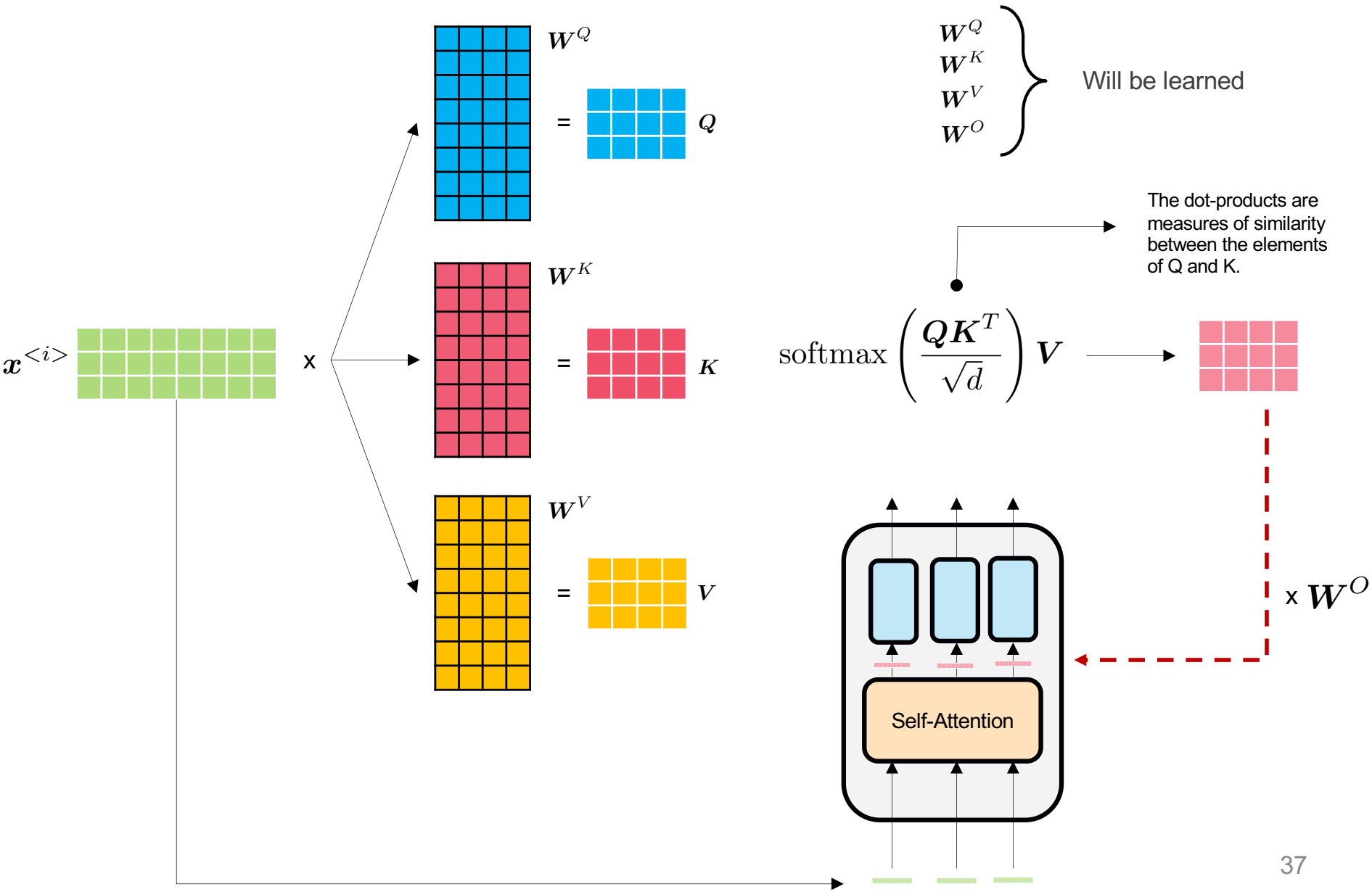
Self-attention: matrix form



Self-attention: matrix form

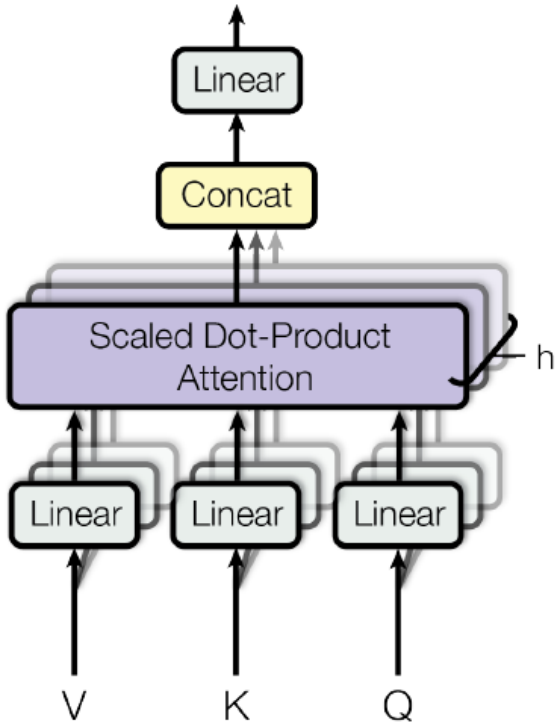


Self-attention: matrix form



Multi-Head Attention

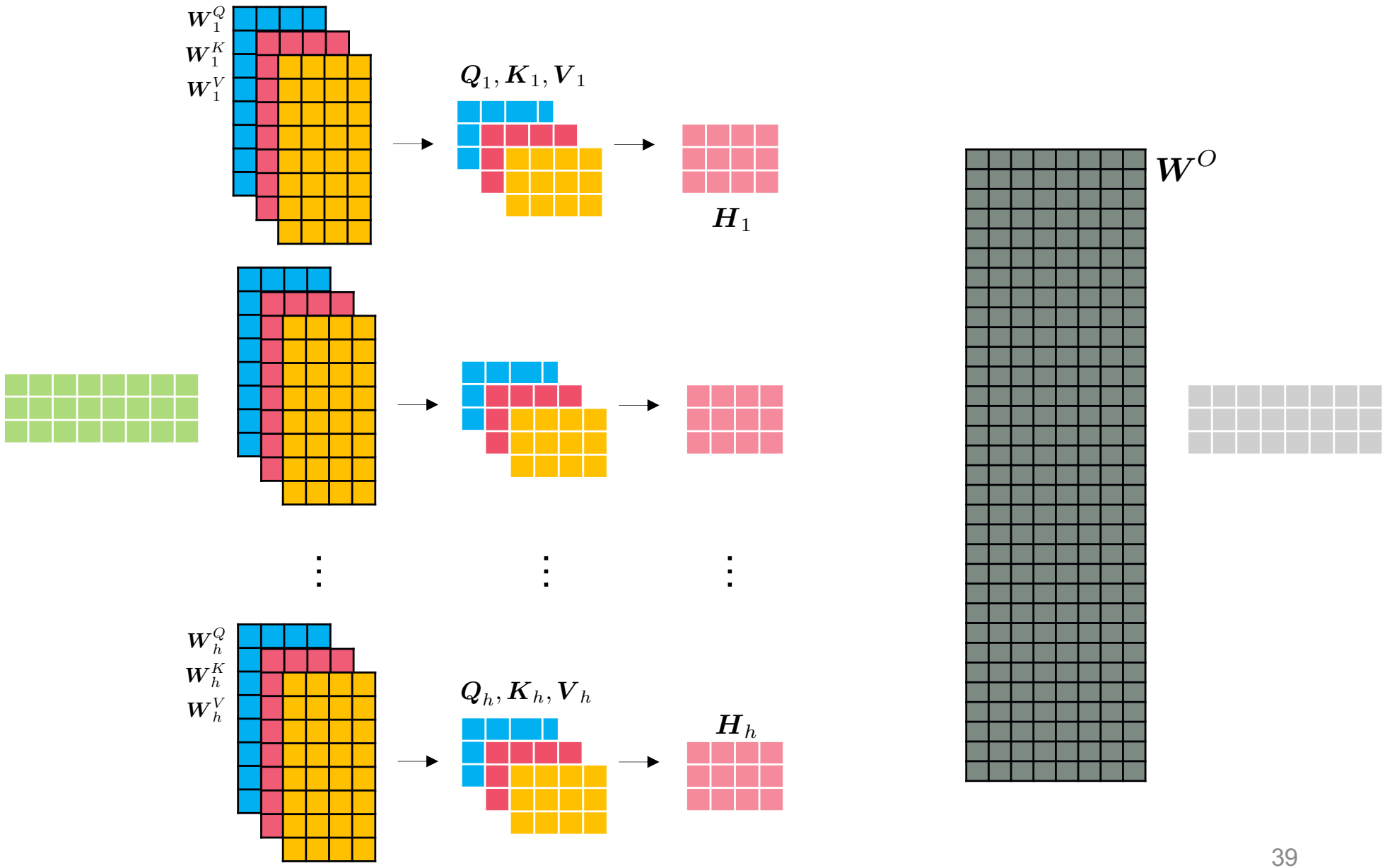
Repeat the previous process *h* times and concatenate the results.



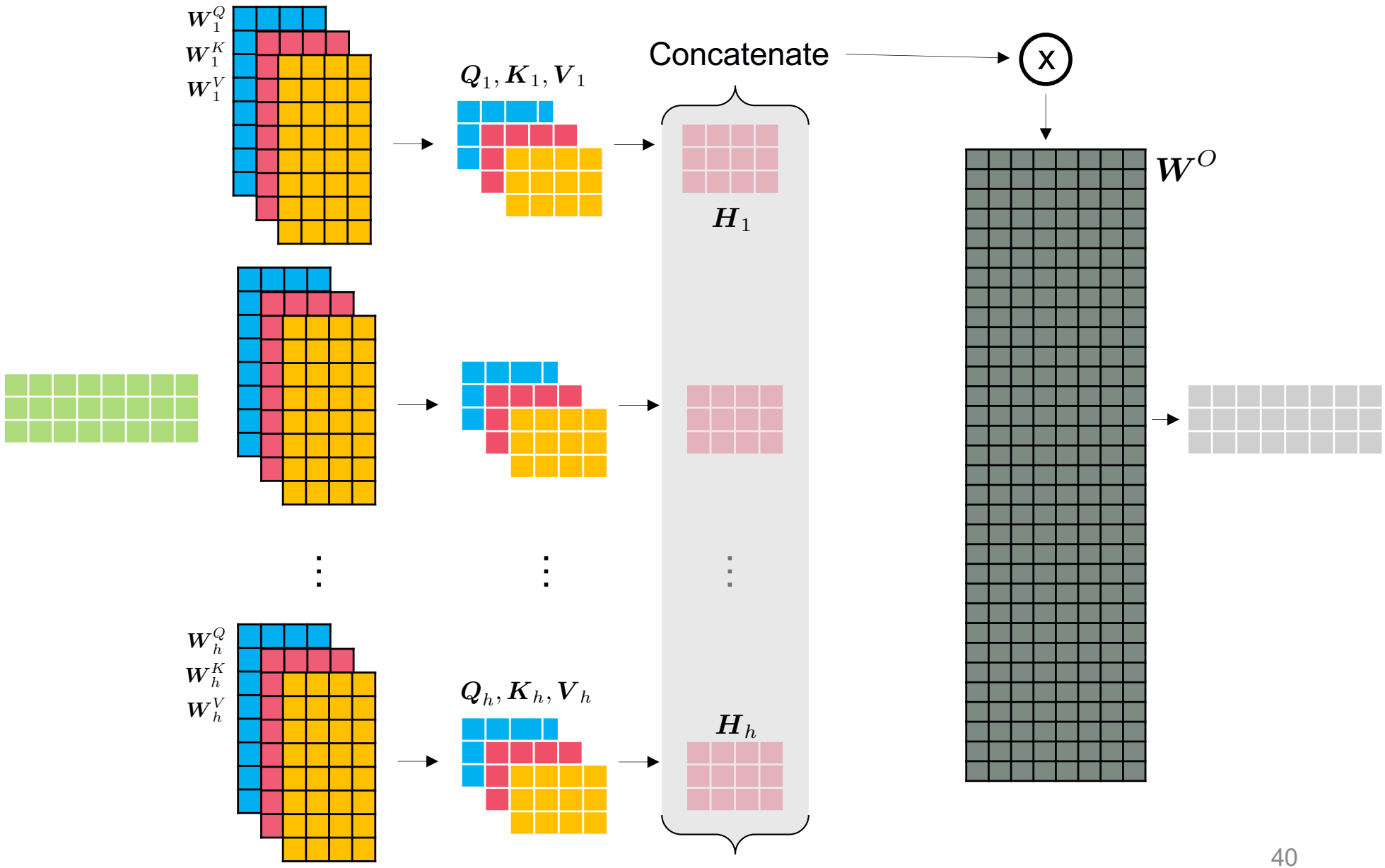
$$\boldsymbol{H}_i = \mathrm{Attention}(\boldsymbol{Q}\boldsymbol{W}_i^Q, \boldsymbol{K}\boldsymbol{W}_i^K, \boldsymbol{V}\boldsymbol{W}_i^V)$$

$$\mathrm{MultiHead}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = [\boldsymbol{H}_1 \cdots \boldsymbol{H}_n]\boldsymbol{W}^O$$

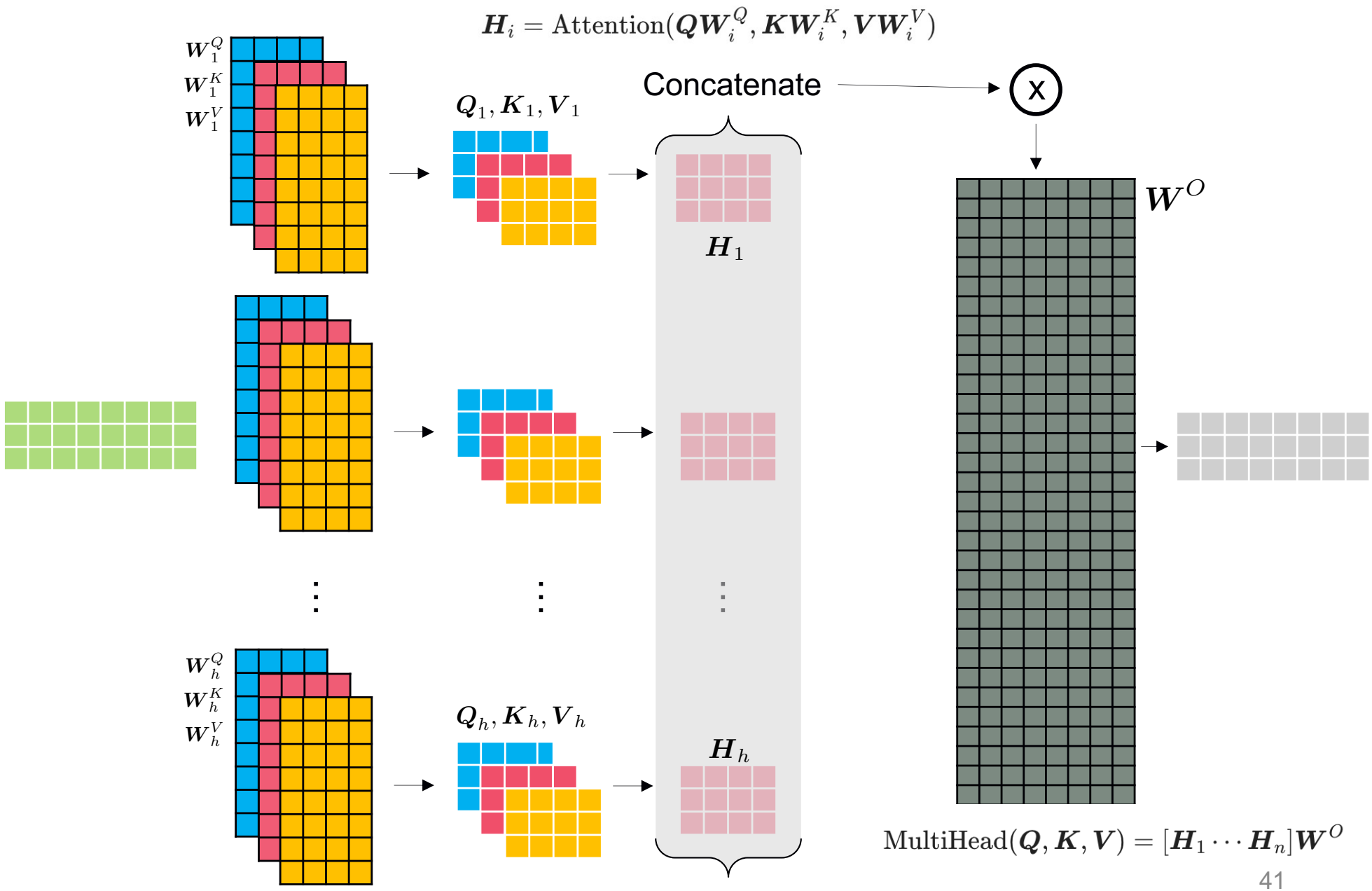
Multi-Head Attention



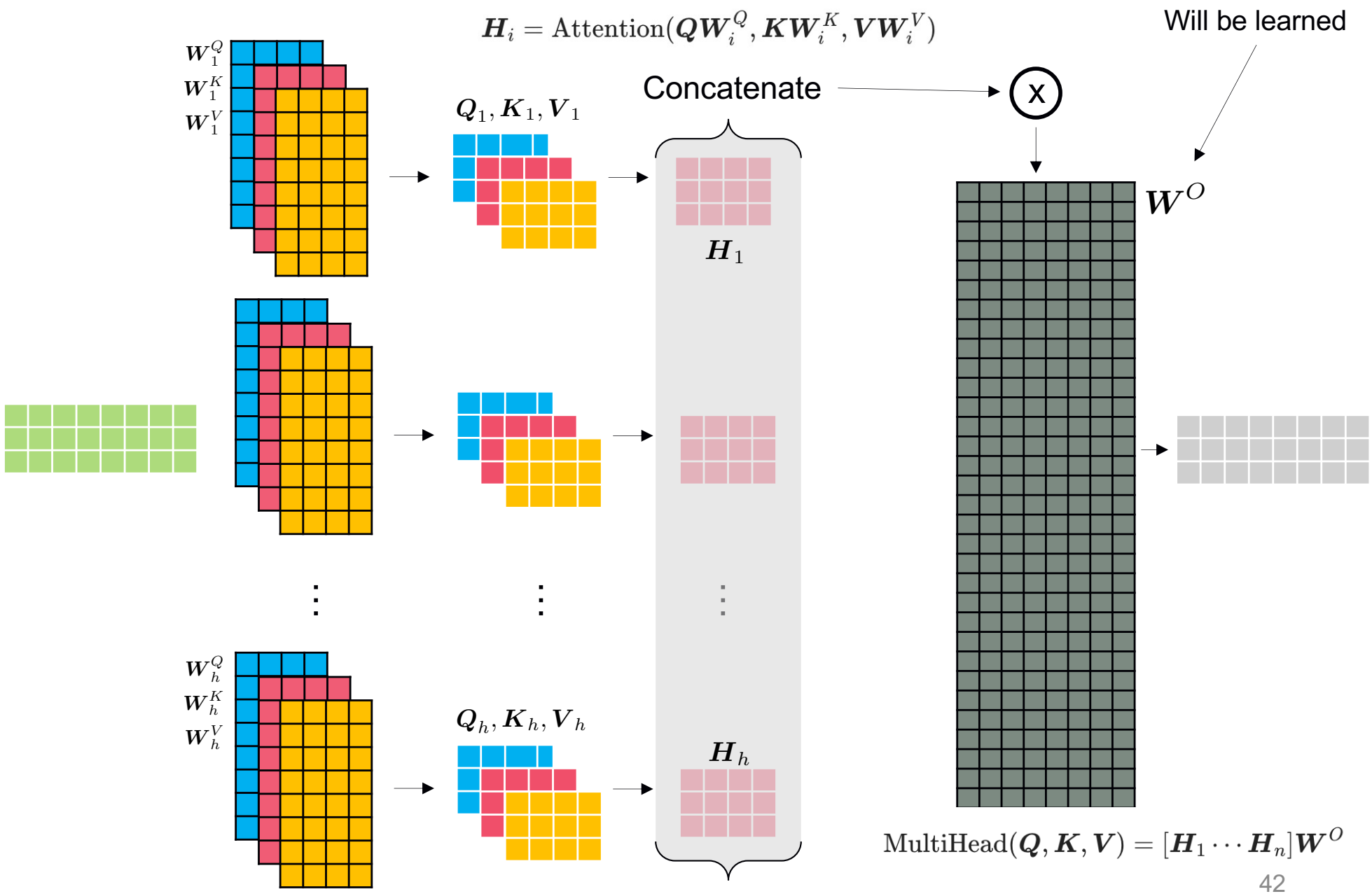
Multi-Head Attention



Multi-Head Attention

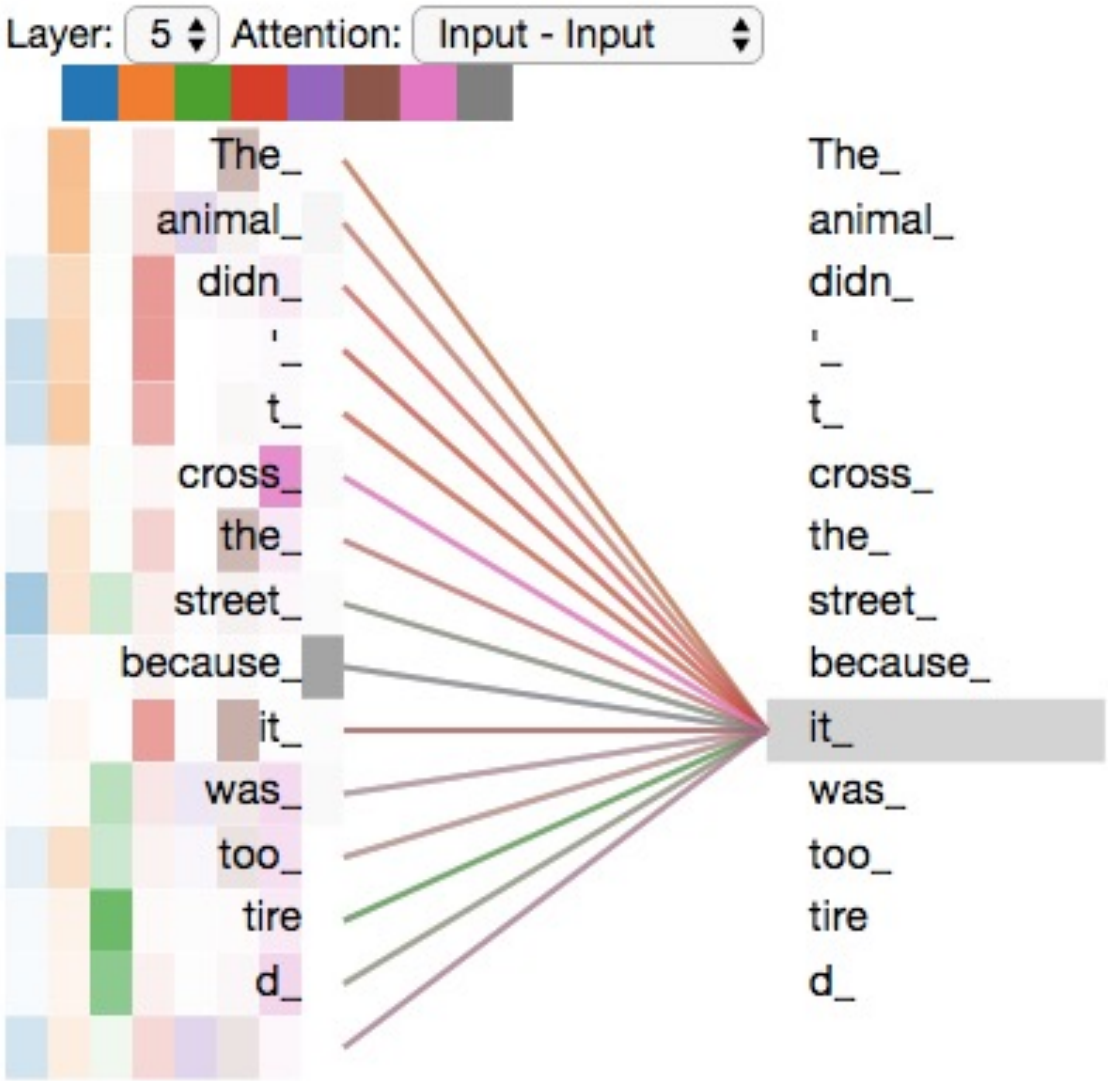


Multi-Head Attention



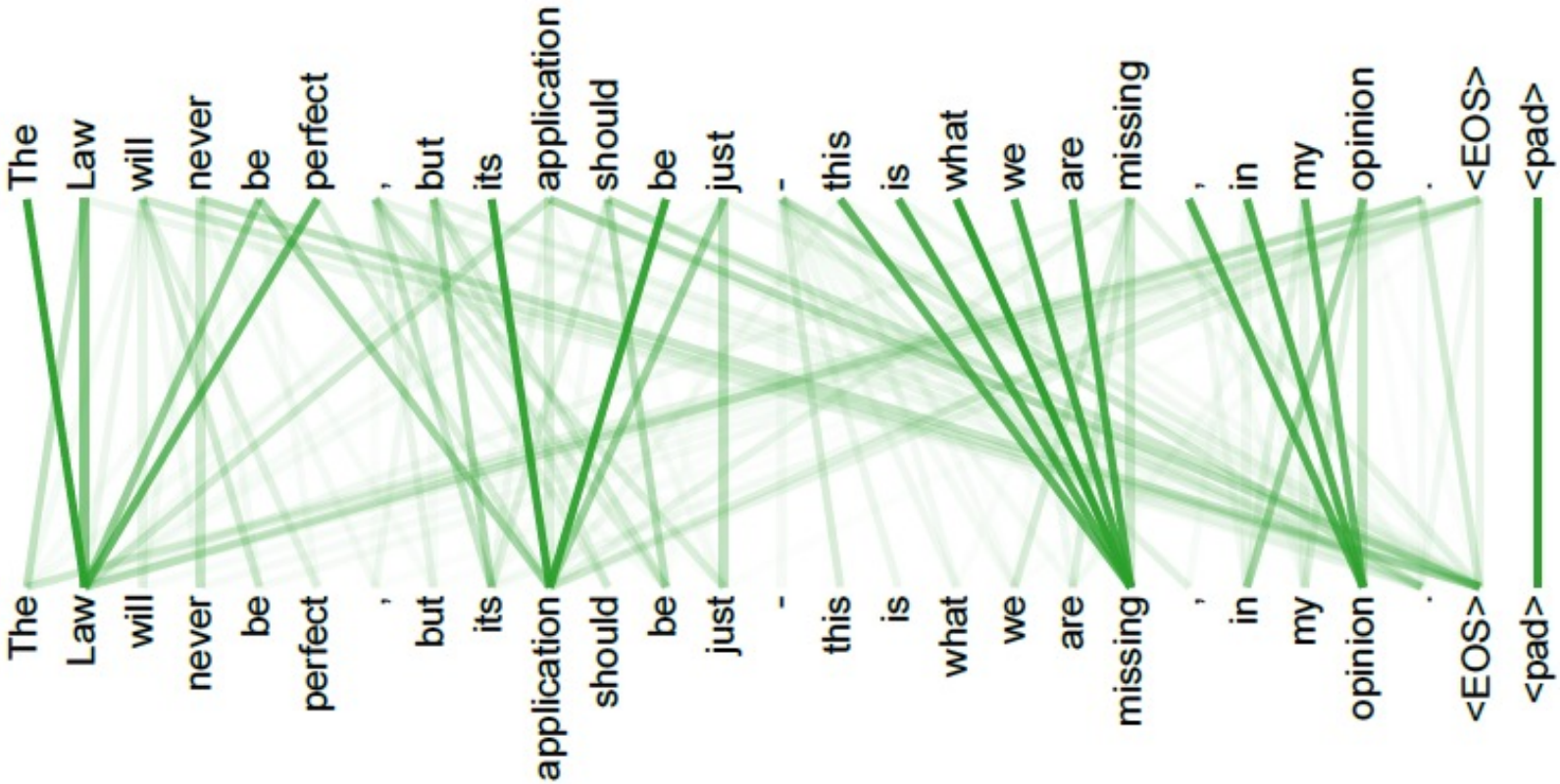
$$\text{MultiHead}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = [\boldsymbol{H}_1 \cdots \boldsymbol{H}_n]\boldsymbol{W}^O$$

Self-attention visualised



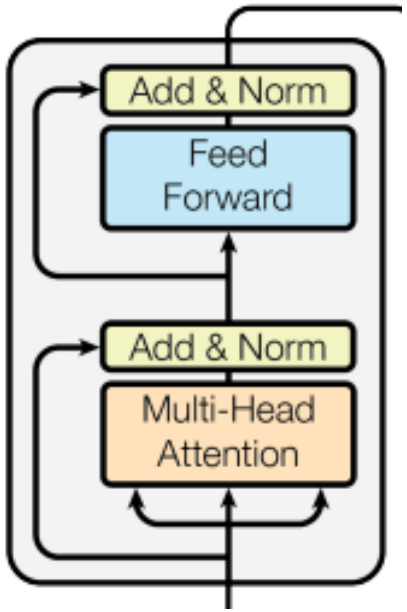
With one attention head one word may dominate.

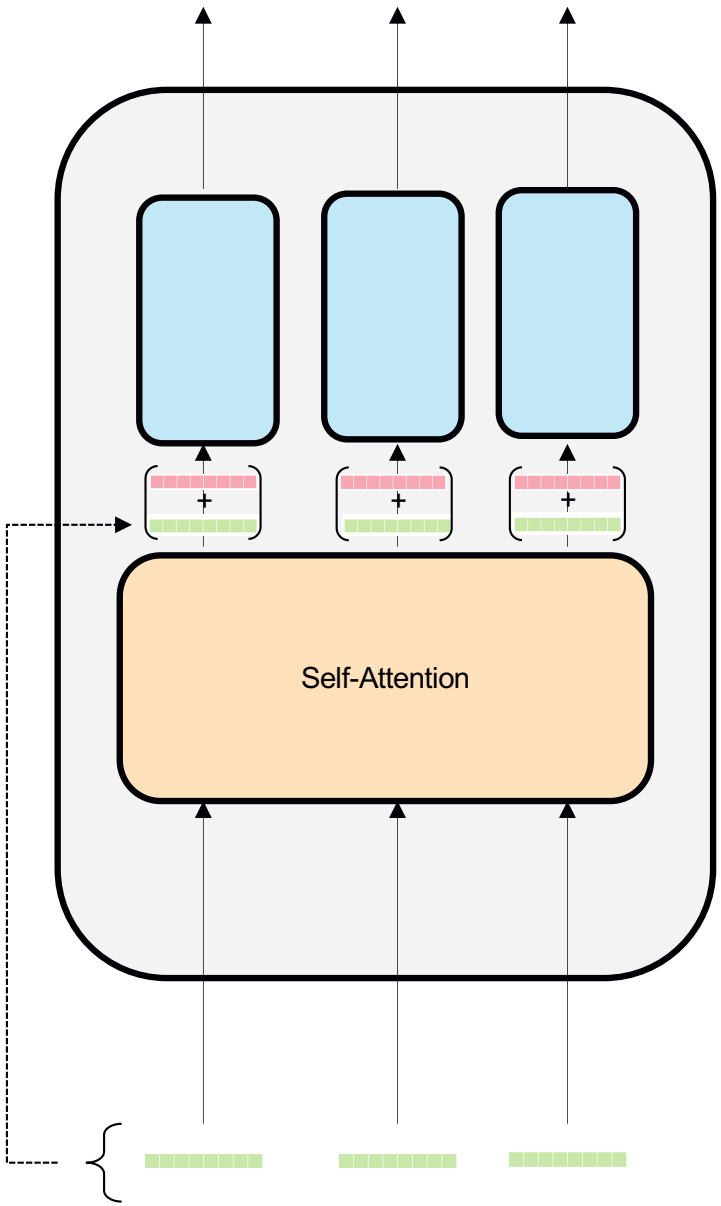
Example from attention layer



From self-attention layer 5 (of 6). Attention seems related to structure of the sentence.

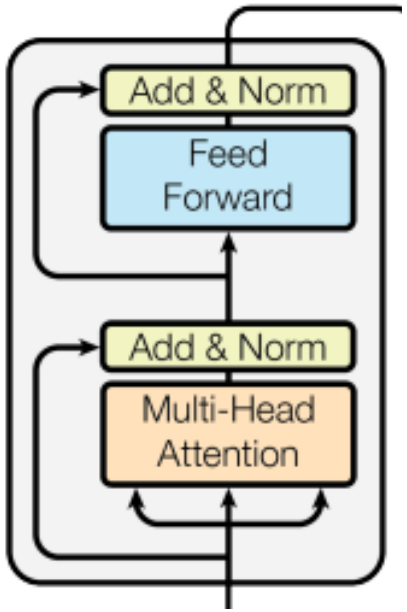
Encoder: Residual connections

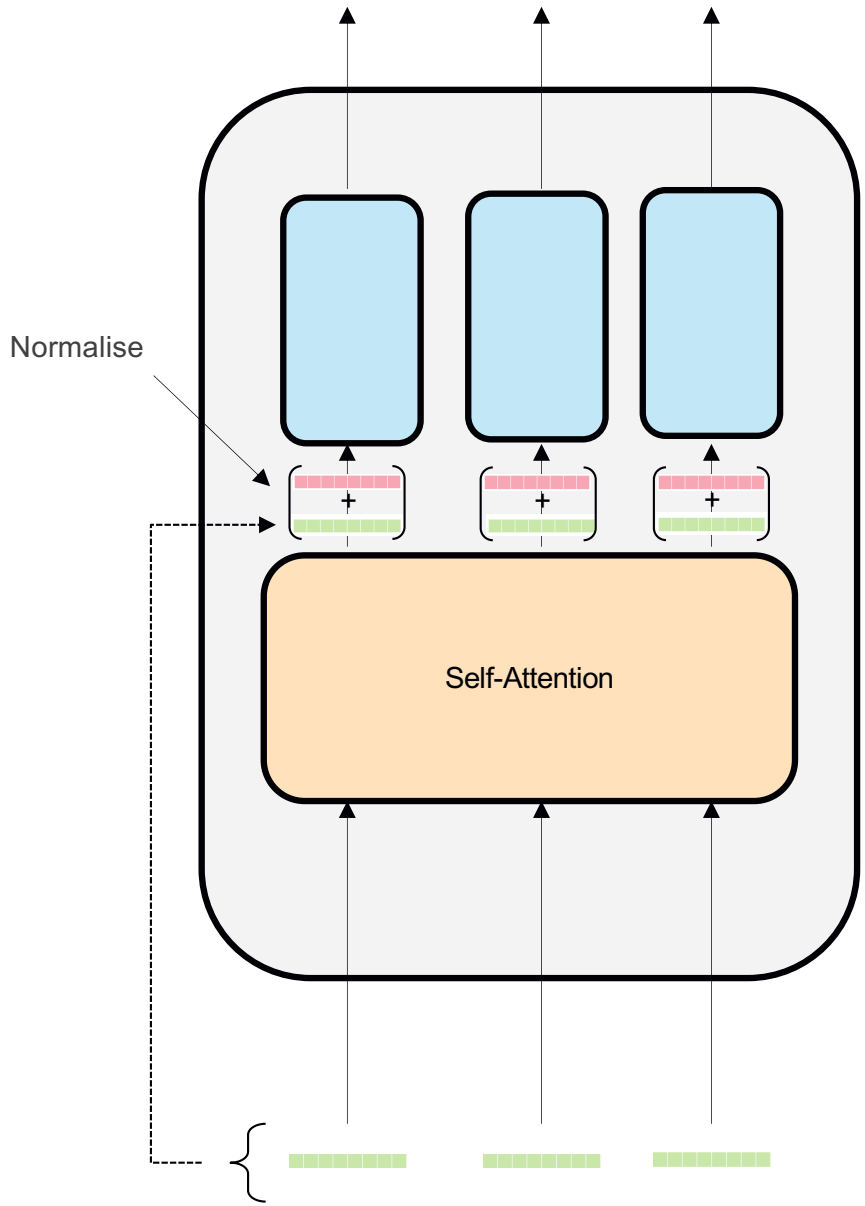




Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Encoder: Residual connections





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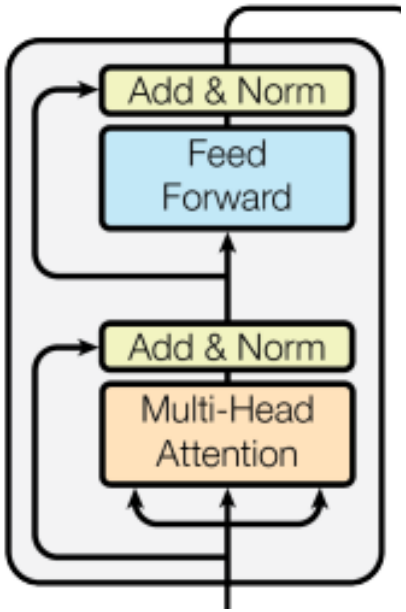
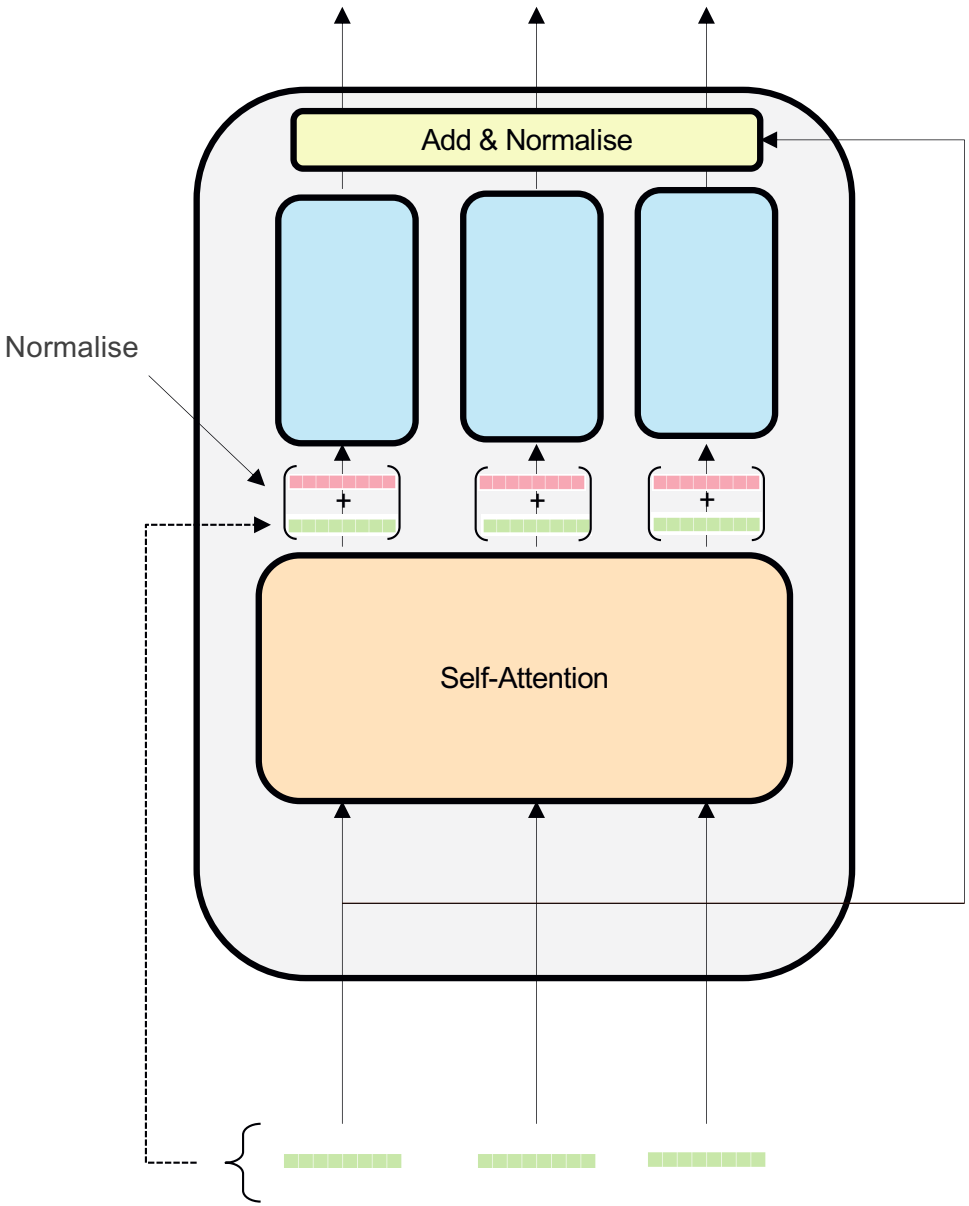
Encoder: Residual connections

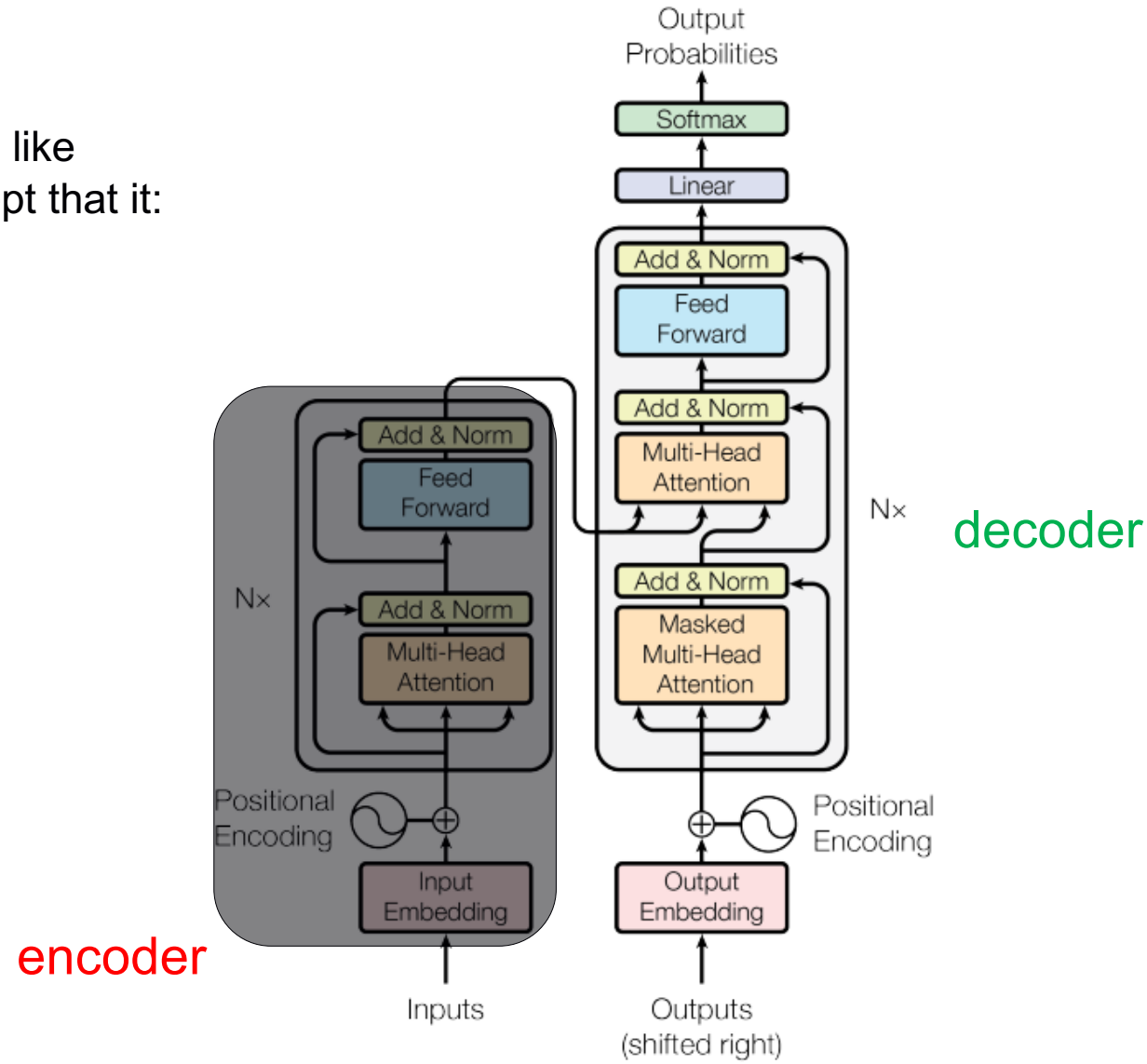




Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Transformer: decoder

- Decoder computations are like those of the encoder, except that it:



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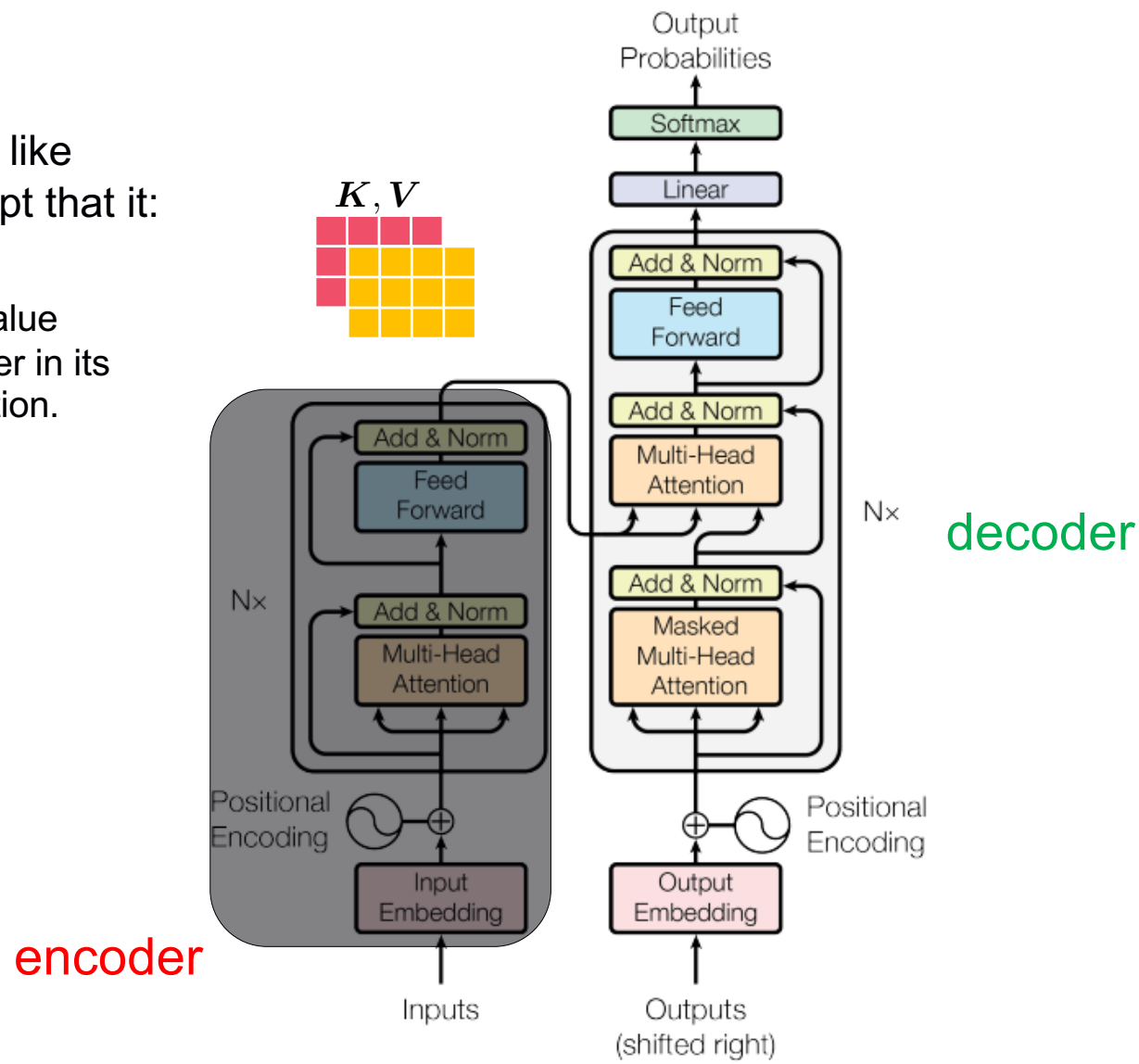
Transformer: decoder

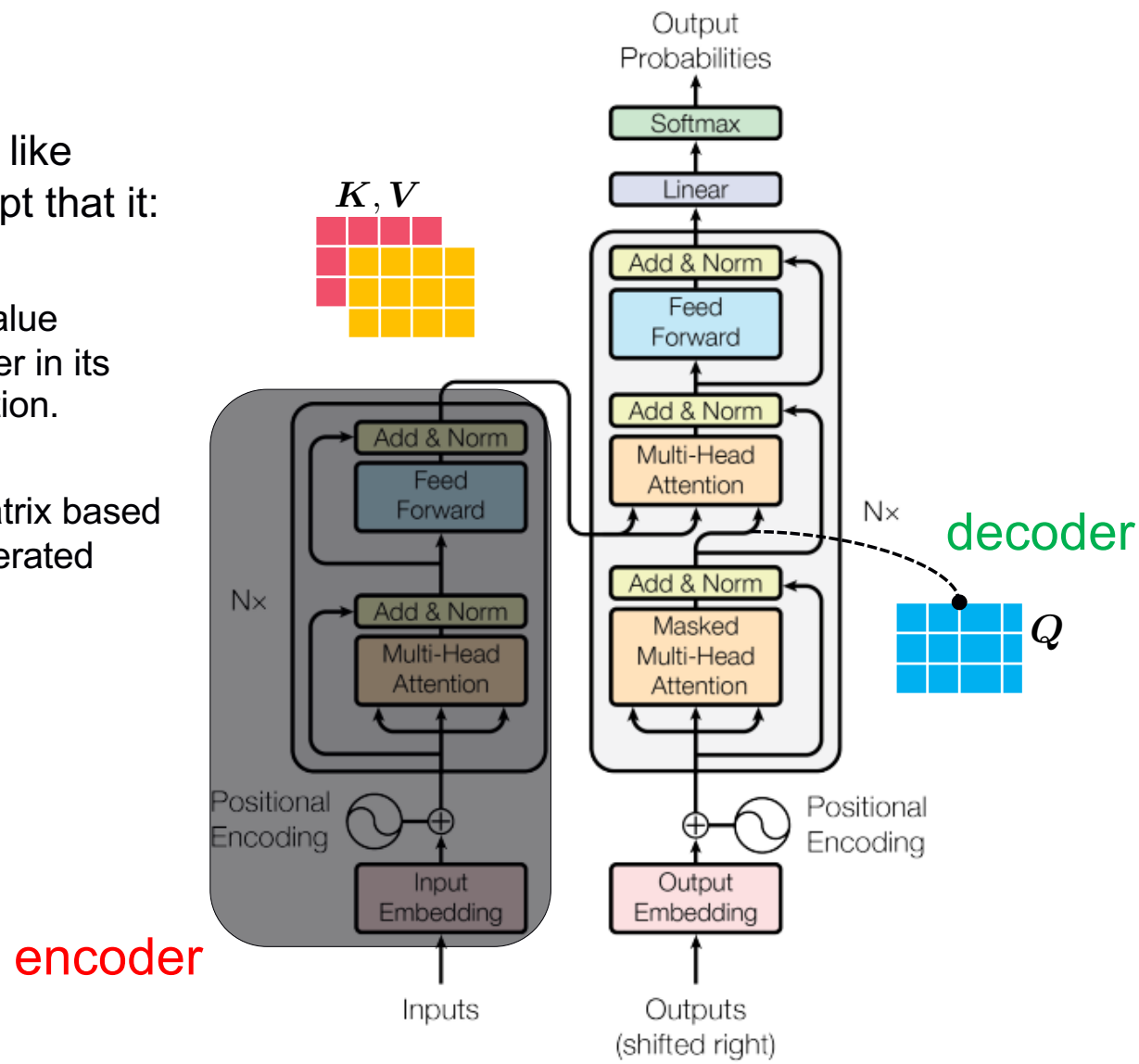
- Decoder computations are like those of the encoder, except that it:
  - Receives the Key and Value matrices from the encoder in its second multi-head attention.



Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Transformer: decoder

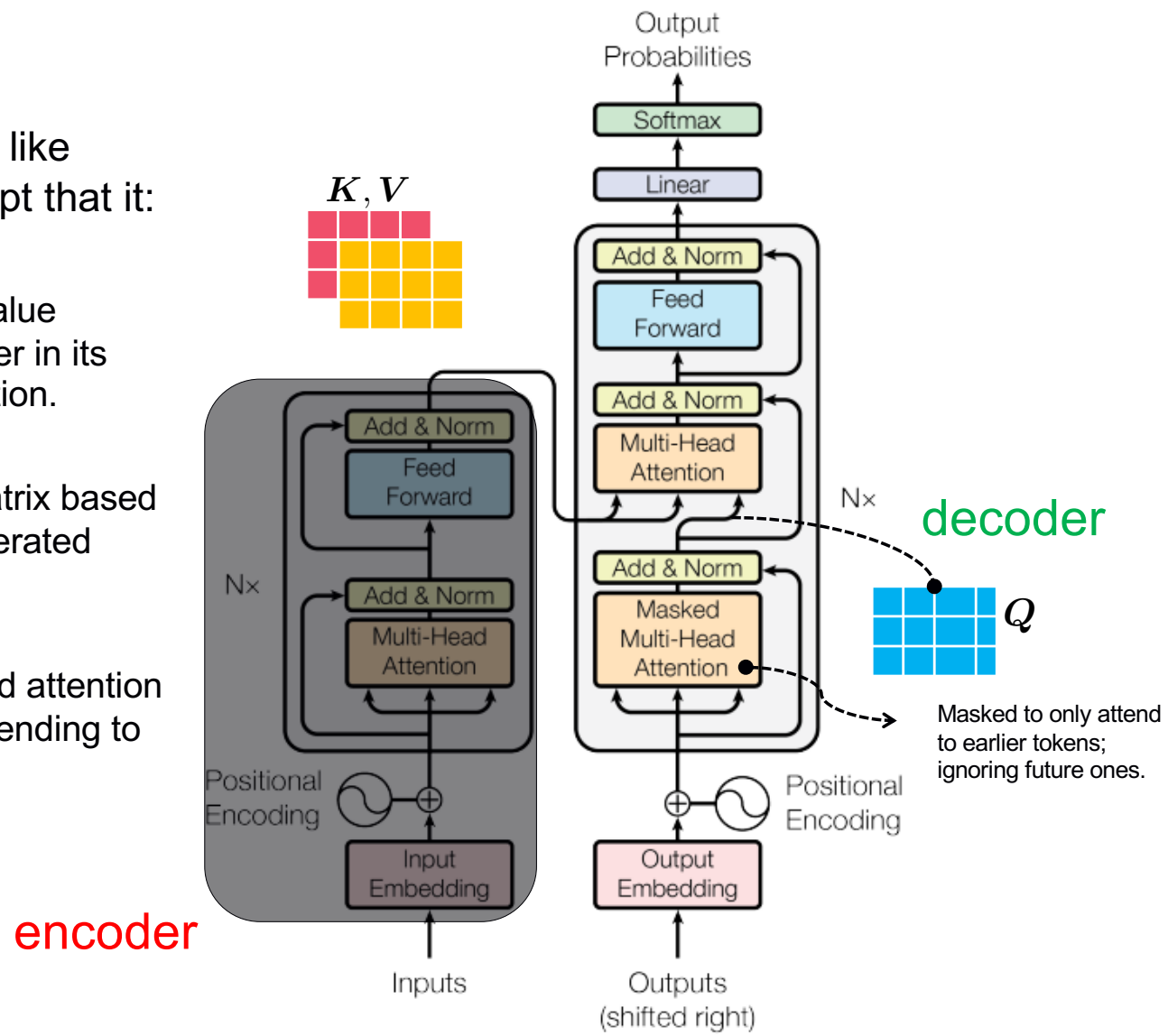
- Decoder computations are like those of the encoder, except that it:
  - Receives the Key and Value matrices from the encoder in its second multi-head attention.
  - Calculates the Query matrix based on its input (i.e., the generated translation tokens).



Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

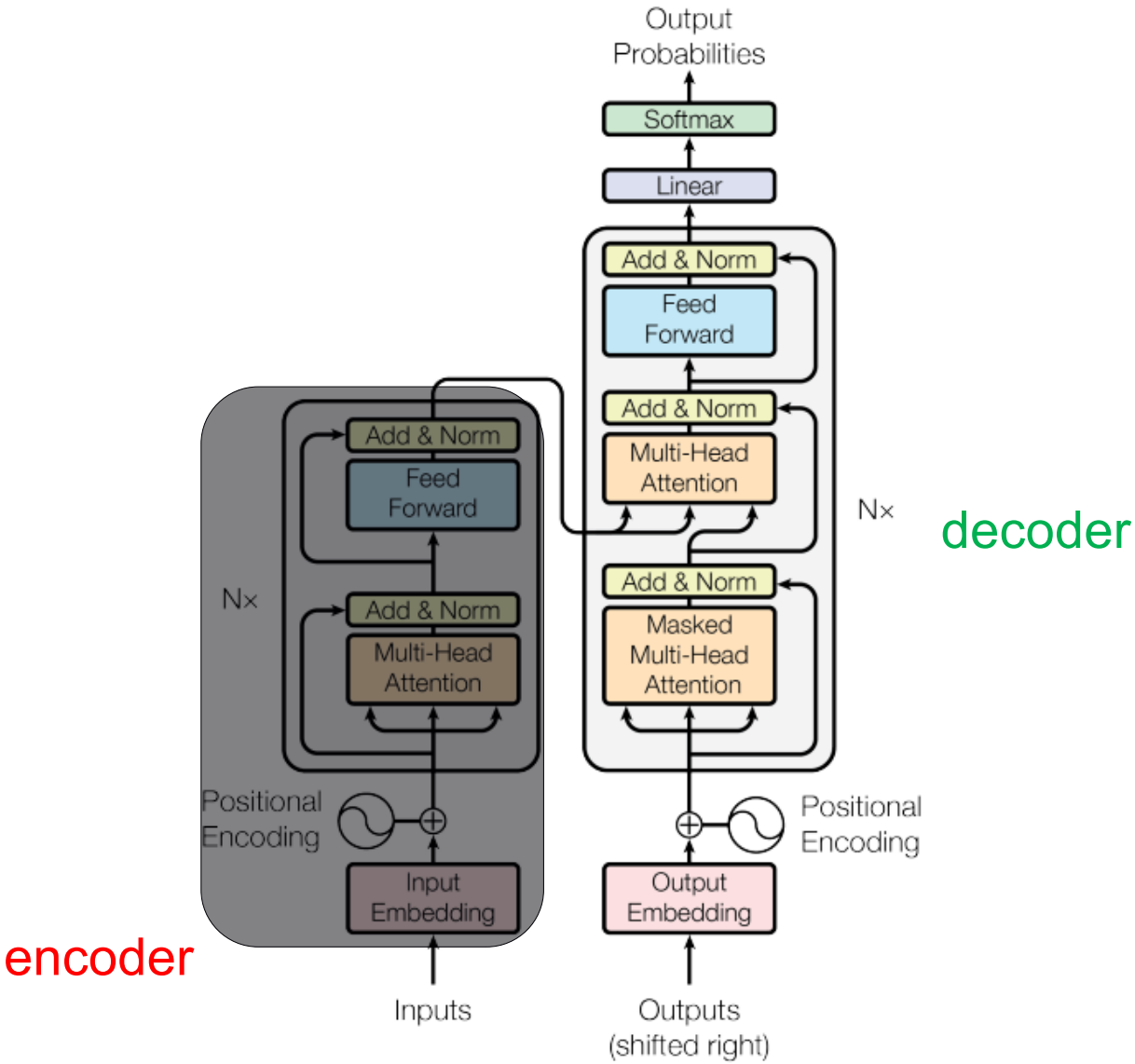
Transformer: decoder

- Decoder computations are like those of the encoder, except that it:
  - Receives the Key and Value matrices from the encoder in its second multi-head attention.
  - Calculates the Query matrix based on its input (i.e., the generated translation tokens).
  - Masks the first multi-head attention unit to prevent it from attending to future positions (tokens).



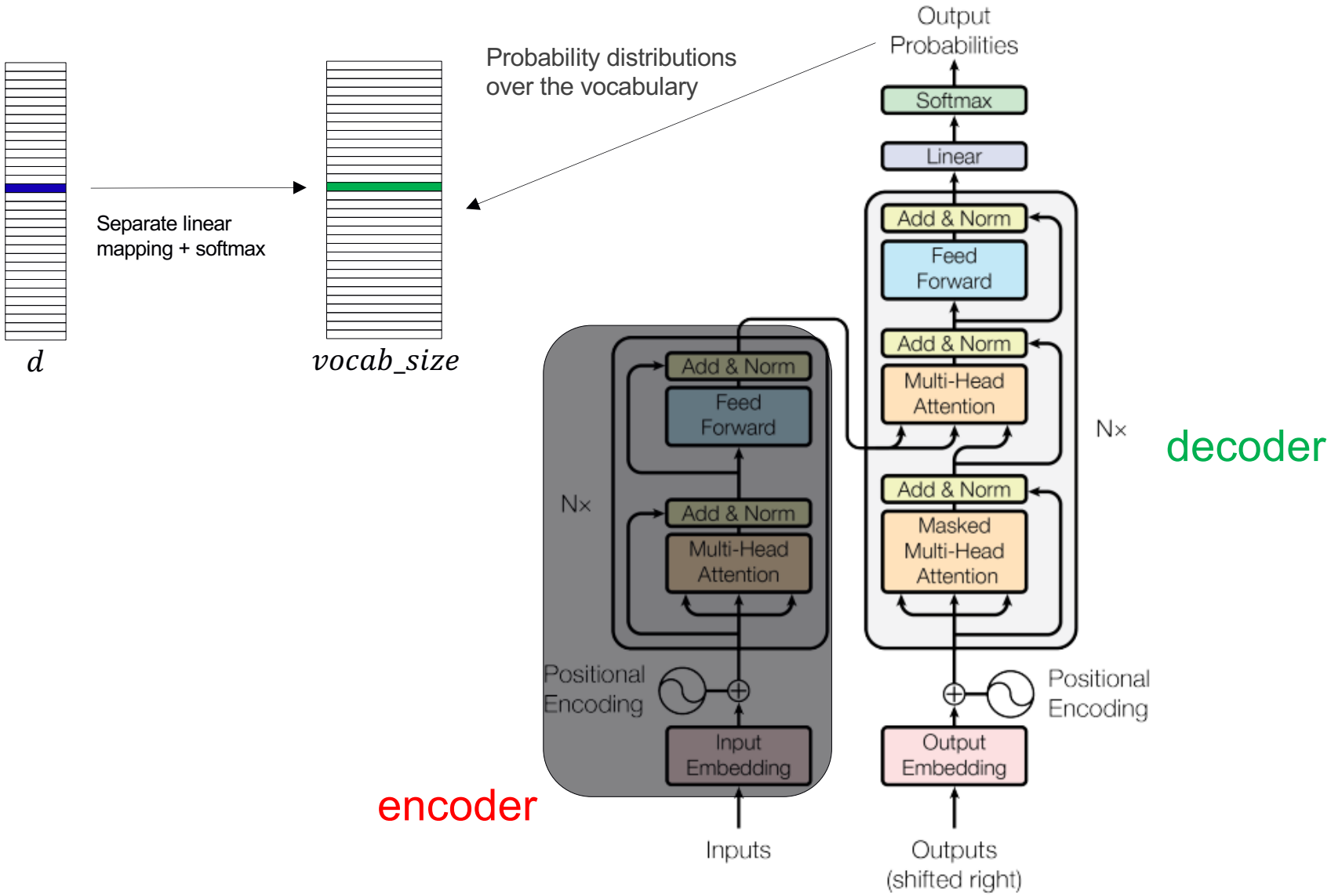
Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Transformer: decoder



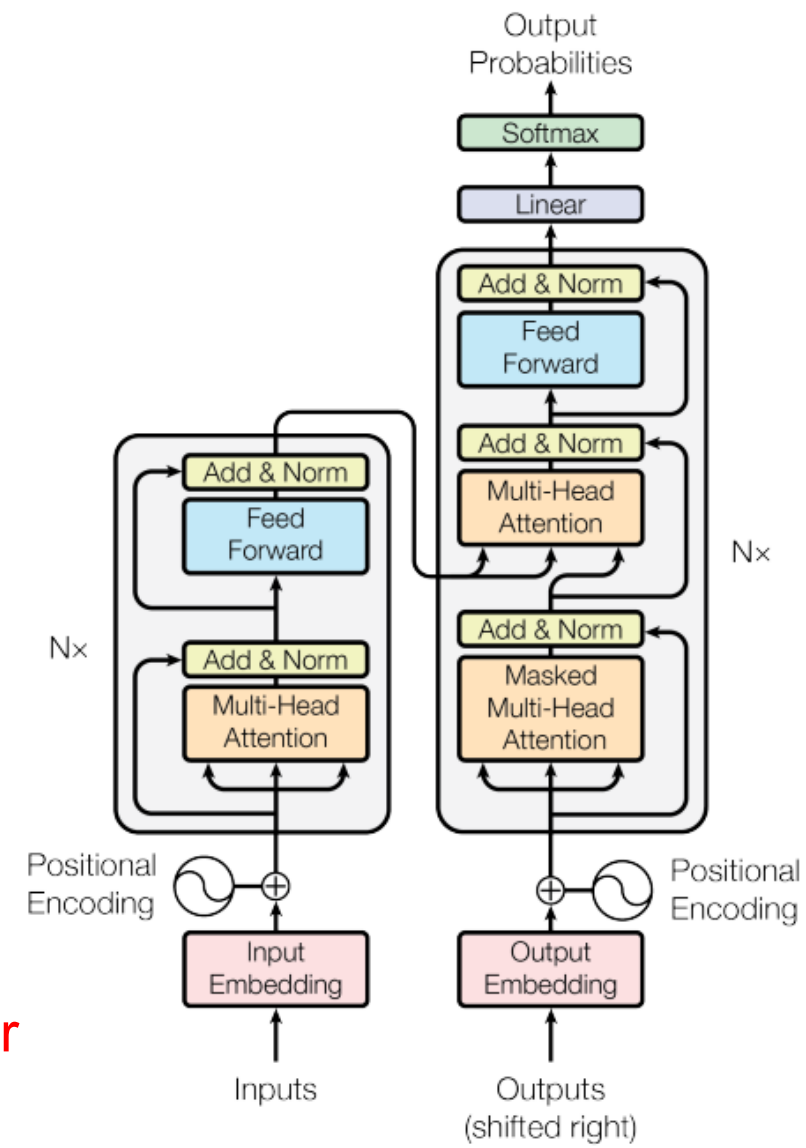
Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Transformer: decoder



Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Transformer

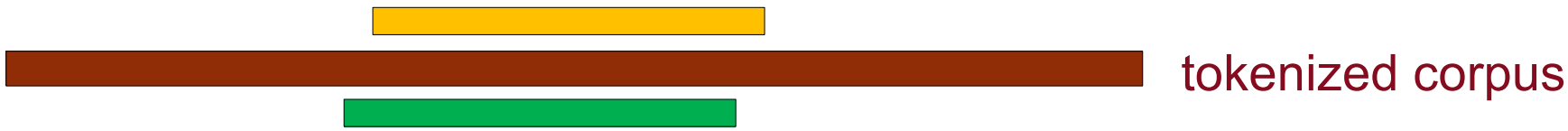


Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

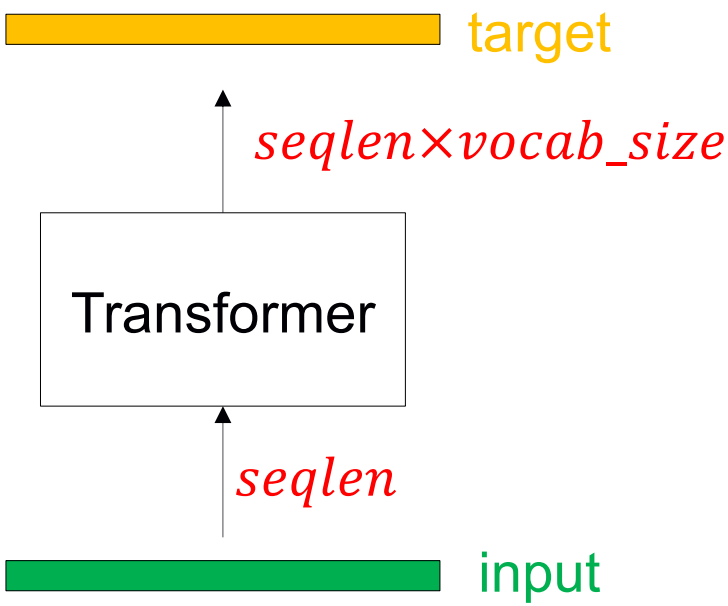
GPT3 – training

From all 2048 positions, model learns to predict the next token

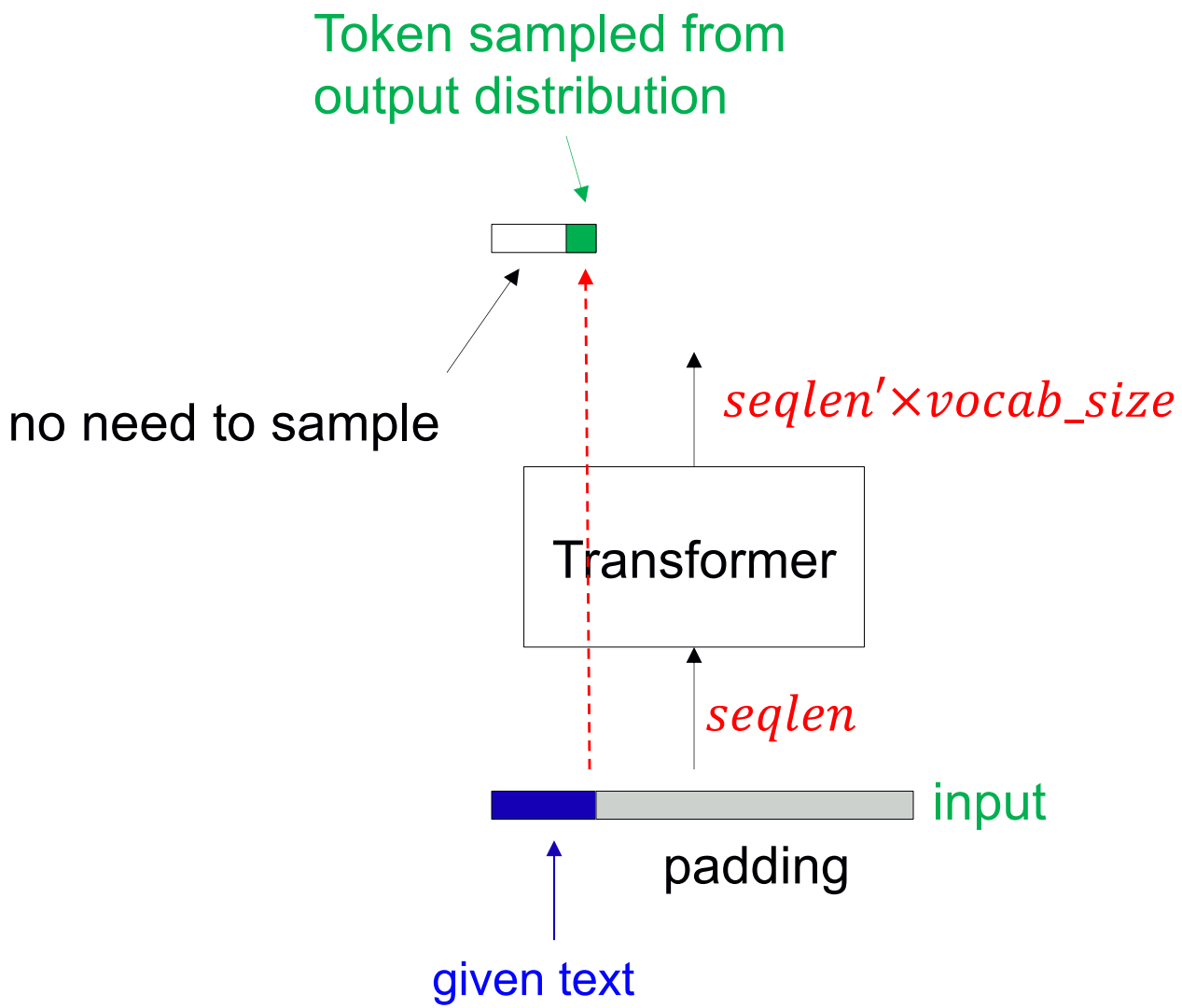
Negative log likelihood (cross-entropy) loss



Target sequence advanced by one token



GPT3 – generation (evaluation mode)



Repeat to extend token by token

When context window full, drop text from the start

GPT3 - news article generation

Title: Star's Tux Promise Draws Megyn Kelly's Sarcasm

Subtitle: Joaquin Phoenix pledged to not change for each awards event

Article: A year ago, Joaquin Phoenix made headlines when he appeared on the red carpet at the Golden Globes wearing a tuxedo with a paper bag over his head that read, "I am a shape-shifter. I can't change the world. I can only change myself." It was a promise to not change to fit into the Hollywood mold: "I think that's a really special thing, to not change yourself. I think it's a really special thing to say, `This is what's inside of me, I'm proud of it, and I'm not going to be ashamed because of the way that someone else thinks I should be.'" Now, it's the Oscars, and Phoenix is at it again. But this time, his publicist is saying he'll be wearing a tux no matter what. Megyn Kelly was not impressed, and she let him have it on The Tonight Show. "You know, I feel like, I feel like you could have worn the tux," she says. "But you're saying you're a shape-shifter. I don't know if you can change your tux, but you can change your mind. You can change your mind. You can change your mind." Phoenix says he did, but it didn't stick. "I was like, `Okay, I'm going to wear a tuxedo to this thing.' And then I thought, `I don't want to wear a tuxedo to this thing.'" Kelly goes on to encourage him to change his mind again, but Phoenix says it's too late: "I'm committed to wearing this."

Language models are few shot learnings, NeurIPS 2020.